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Weight-level integration of SHAP-interpreted machine learning and AHP for flood susceptibility mapping in a semi-arid region

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ABSTRACT

Flood susceptibility mapping is critical for urban planning in semi-arid regions exposed to high-intensity seasonal rainfall and rapid urban growth. Conventional approaches often involve subjective weighting and limited interpretability. This study proposes a GIS-based weight-level integration approach linking interpretable machine learning with the Analytic Hierarchy Process (AHP). Four tree-based models were trained using standard and spatial cross-validation; SHapley Additive exPlanations (SHAP) quantified factor contributions, enabling recalibration of expert-derived weights. CatBoost demonstrated the strongest generalization, achieving 85.19% standard and 73.30% spatial accuracy. ROC-AUC analysis showed that AHP + SHAP and standalone AHP attained the highest discriminative performance (AUC = 0.964), outperforming conventional map-level integration (AUC = 0.958) and standalone CatBoost (AUC = 0.859). The framework proved robust to weight perturbations, unlike map-level integration which produces polarized patterns. Weight-level integration yields balanced spatial distributions and reduces false alarms, providing a transparent scheme for data-constrained urban contexts to support informed environmental decision-making.

1. Introduction

Floods are among the most destructive natural hazards worldwide, causing substantial human, economic, and environmental losses (Manzoor et al., 2022; Merz et al., 2021; Yu et al., 2022). They recurrently lead to fatalities, damage to infrastructure, disruption of livelihoods, and long-term socio-economic vulnerability, particularly in flood-prone and rapidly urbanizing regions (Chebika et al., 2025; Guoyi et al., 2023; Rentschler et al., 2023). Recent studies report a continuous increase in flood-related losses, largely driven by climate change, the intensification of extreme precipitation events, and unplanned urban expansion (Bibi and Kara, 2023; Cea and Costabile, 2022). Consequently, developing reliable and decision-oriented flood susceptibility assessment frameworks has become a critical priority for effective flood risk reduction and sustainable urban planning.

Over the past decades, advances in geospatial technologies and spatial decision support systems, particularly Geographic Information Systems (GIS) and remote sensing, have substantially improved flood susceptibility mapping by enabling the acquisition, integration, and analysis of spatial data over large areas with high spatial and temporal resolution (Mashala et al., 2023; Pei et al., 2021; Zafarmomen et al., 2024). Satellite-based observations facilitate continuous monitoring of hydrological and environmental processes and support the extraction of multiple flood-conditioning factors, including topography, land use and land cover, drainage networks, and rainfall characteristics (Ahmad et al., 2025; Taherizadeh et al., 2023; Uchua et al., 2025). These developments have established GIS-based environments as the primary analytical platform for flood susceptibility analysis and spatial decision-making.

With the increasing availability of geospatial data, flood susceptibility mapping has become a central component of flood risk management and spatial planning. Early studies predominantly relied on physically based hydrological and hydraulic models, which, despite their strong theoretical foundations, require extensive input data,

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complex calibration procedures, and high computational demands. These requirements limit their practical applicability at regional and large spatial scales (Kumar et al., 2023). To address these constraints and to better exploit multi-source geospatial information within GIS environments, expert-based approaches and multi-criteria decision analysis methods have been widely adopted. multi-criteria decision analysis (MCDA) Methods offer flexibility and computational efficiency and allow the explicit integration of multiple flood-conditioning factors through weighted aggregation (Aldarwish et al., 2026; Kader et al., 2024; Kaya and Derin, 2023). However, a fundamental limitation of MCDA-based flood susceptibility mapping lies in its strong dependence on expert judgment, as weight assignments inevitably reflect individual perceptions, experience, and potential biases, introducing subjectivity and uncertainty that affect the reliability and reproducibility of the resulting susceptibility maps (Islam et al., 2025).

In recent years, ML and deep learning (DL) techniques have emerged as powerful alternatives for flood susceptibility mapping, with numerous studies demonstrating improved predictive performance compared to traditional expert-driven approaches (Akinsoji et al., 2025; Huang et al., 2025; Nachappa et al., 2020; Tarek et al., 2023). These data-driven methods are particularly effective in capturing complex and nonlinear relationships among flood-conditioning factors while reducing direct human bias during model development (Abdo et al., 2025; Islam et al., 2025; Saberian et al., 2026; Zafarmomen et al., 2026). Furthermore, recent advances in model interpretability have enhanced the transparency and applicability of ML-based models in decision-support contexts. In this regard, SHapley Additive exPlanations (SHAP) has been increasingly employed to quantify the contribution of individual predictors to model outputs, partially addressing the black-box nature of many ML algorithms (Aydin and Iban, 2023; Chu et al., 2026; Y. Wang et al., 2025; Zerouali et al., 2025).

Despite these advances, ML and DL models continue to face important limitations. Their performance depends strongly on the availability of large and representative training datasets, and predictive reliability may decrease when applied to spatially heterogeneous environments or conditions not adequately represented during training (Al-Emadi et al., 2025; Meyer et al., 2019). Moreover, even with interpretability tools such as SHAP, complex models may remain difficult to fully understand, potentially leading to misinterpretation or overreliance by decision-makers (Diehr et al., 2025; Dikshit and Pradhan, 2021). An exclusive reliance on purely data-driven approaches may also underutilize valuable expert knowledge, highlighting the need for integrative frameworks that balance predictive accuracy with domain-informed reasoning (Albrecht et al., 2020; El-Maissi et al., 2024).

Against this background, a critical methodological gap remains unresolved in flood susceptibility research. Most existing studies either compare MCDA and ML approaches or combine their outputs at the final map level, without addressing integration at the feature importance or weighting stage (Adeyemi et al., 2024; Bui et al., 2023; Nachappa et al., 2020; Shikhteymour et al., 2023). Consequently, MCDA Methods remain dependent on subjective expert-assigned weights, while ML models operate as independent predictive tools, limiting the effective transfer of data-driven insights into decision-oriented flood susceptibility mapping. Although SHAP has been widely used to interpret ML models, its outputs are typically employed for post hoc explanation rather than being systematically incorporated into MCDA weighting schemes.

To address these challenges, this study develops an integrative framework that links interpretable ML with MCDA, implemented through the AHP, at the weighting stage. Multiple tree-based ML models are trained using historical flood and non-flood data, and SHAP values are employed to derive data-driven estimates of the relative contribution of conditioning factors. The normalized SHAP-derived importance scores extracted from four ML models are aggregated and subsequently integrated directly with the expert-derived AHP weights. The framework is examined through several comparative scenarios, including standalone ML using the best-performing model, a conventional AHP approach based solely on expert judgment, map-level integration of ML and AHP outputs, and the proposed weight-level integration framework, allowing for a careful assessment of the potential benefits of incorporating ML interpretability into MCDA-based flood susceptibility mapping. The proposed methodology is applied to M'sila, a semi-arid urban environment in North Africa characterized by pronounced rainfall variability, episodic high-intensity precipitation events, and rapid urban expansion. The study area exhibits hydro-climatic and urban characteristics typical of semi-arid North African cities, providing a representative and rigorous test case for evaluating the effectiveness and applicability of the proposed integrated approach.

In summary, the primary contribution and novelty of this study lie in the development of a weight-level integration framework that combines interpretable machine learning (SHAP) with expert-based multi-criteria decision analysis (AHP). This is achieved by aligning the mean SHAP values derived from multiple ML models with expert-derived weights within a unified structure, rather than relying on conventional map-level integration. Furthermore, the study provides a systematic comparative evaluation of multiple modelling configurations to demonstrate the added value of the proposed framework.

2. Materials and methods

2.1. Study area

The semi-arid city of M'sila in Algeria, North Africa, located between 4.48°E and 4.57°E longitude and 35.66°N and 35.76°N latitude (see Fig. 1), was selected as the study area for the application of the proposed approach. The city was chosen due to its regional significance, the recurrent occurrence of floods (Chouki and Makhloufi, 2022; Loumi and Ali, 2021), and the research team's in-depth knowledge of the area, which facilitated the compilation of a reliable dataset. Furthermore, flood-related studies in M'sila are limited and have primarily relied on traditional methodologies, highlighting the value of generating up-to-date and robust information to support decision-makers and emergency response units prior to flood events.

2.2. Overall methodology

This study proposes an integrated framework that combines interpretable ML and MCDA to produce more transparent flood susceptibility maps using a spatially standardized set of conditioning factors. Four supervised models were trained and evaluated, and SHAP were employed to derive data-driven factor weights. The best-performing model was applied to generate a machine learning–based flood susceptibility map. In parallel, an expert-driven MCDA map was developed using the AHP. Two integration scenarios were then implemented: (i) map-level integration, combining the expert-based map with the output of the best-performing ML model, and (ii) weight-level integration, in which SHAP-derived importance was systematically combined with expert-assigned weights within the AHP framework, followed by weighted overlay analysis. Both scenarios integrate expert knowledge and data-driven information to support interpretable flood susceptibility mapping. The overall workflow is summarized in Fig. 2.

2.3. Data preparation

We selected seven primary conditioning factors based on previous studies and local knowledge of the study area (Islam et al., 2025; Kaya and Derin, 2023). A 30-m USGS DEM was used to derive topographic features (Slope, Elevation, Plan Curvature, Depth of Depressions) and Distance to the Drainage Network. Land use/land cover data for 2024 were obtained from ESRI Land Cover, and precipitation points from POWER NASA were interpolated using IDW. All layers were standardized to the DEM resolution. Each factor contributes to flood susceptibility in a specific way: Slope affects surface runoff and water

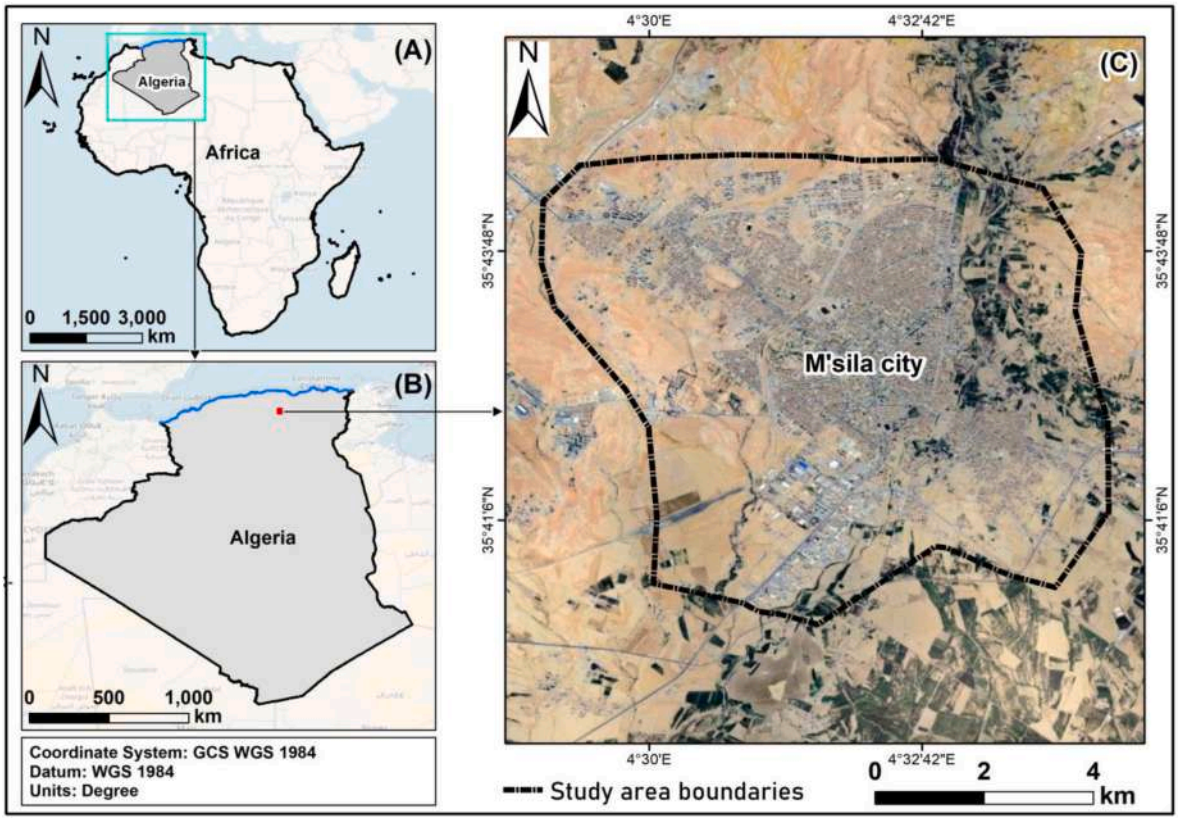


Fig. 1. Study Area Map: (a) Algeria in Africa, (b) Study Area in Algeria, (c) Study Area (M'sila city).

accumulation; Elevation identifies low-lying areas prone to flooding; Plan Curvature influences flow convergence or divergence; Depth of Depressions indicates potential water storage areas; Distance to the Drainage Network reflects proximity to channels that can convey or retain water; Land use/land cover represents surface permeability and runoff potential; and Precipitation drives flood events.

To link these factors with actual flood occurrences, flooded and non-flooded points were collected from Landsat imagery, historical records, and reports. Landsat images were specifically reviewed to verify visible water accumulation one to two days after reported floods. Non-flooded points were selected based on local studies that classify the area as non-flood-prone or low-risk areas, as well as areas with no recorded historical flood events or no reported occurrences. All points were cross-checked with local specialists. Selected historical flood events are presented in Appendix A (Fig A1 and A2) as supporting figures and visual materials. After removing three duplicates, the final dataset included 150 flooded and 140 non-flooded points, providing a clean dataset for the development and comparative evaluation of the models. Fig. 3 shows the spatial distribution of the seven conditioning factors along with the locations of flooded and non-flooded points.

2.4. Machine learning-based flood susceptibility mapping

After data preparation, a training dataset was constructed using seven conditioning factors as input features and a binary target variable representing flood occurrence (1 = flood, 0 = non-flood). The dataset was checked for missing values and duplicate records to ensure data quality, and multicollinearity among predictor variables was assessed using correlation analysis. Global Moran's I was computed within a GIS environment to evaluate the spatial autocorrelation of the full binary training dataset, and the corresponding results are provided in Appendix B (Fig B1). All subsequent preprocessing steps, including transformation of land use/land cover data into binary variables using one-hot encoding, as well as model development, training, and evaluation of multiple ML models, were implemented in Google Colab using Python-based libraries.

Four supervised ML algorithms (XGBoost, CatBoost, LightGBM, and Random Forest) were selected due to their proven effectiveness in flood susceptibility mapping and their compatibility with SHAP (Aydin and Iban, 2023; M. Wang et al., 2023; Zerouali et al., 2025). The predictive models were trained and evaluated using both stratified and spatial cross-validation schemes with three folds, which was deemed appropriate given the relatively limited sample size while ensuring sufficient representation within each fold. The use of spatial cross-validation was motivated by the significant spatial autocorrelation detected in the dataset, as indicated by Moran's I ($I = 0.766$, $z = 16.15$, $p < 0.001$), confirming strong positive spatial dependence among observations. Accordingly, spatial cross-validation was adopted as the primary evaluation strategy to provide a more realistic assessment of model generalization to unseen spatial locations. A stratified cross-validation scheme was additionally employed as a non-spatial baseline, preserving the proportion of flooded ($n = 150$) and non-flooded ($n = 140$) samples across folds. This parallel setup enables quantification of the extent to which model performance may be overestimated when spatial dependence is not explicitly considered.

For spatial validation, a block-based approach was applied by partitioning the study area into three geographically distinct zones. The partitioning ensured a balanced representation of both classes within each spatial fold while preserving the original spatial locations of the samples. A map illustrating the spatial partitioning and fold structure is provided in Appendix B (Fig B2). All feature preprocessing steps were performed independently within each fold to prevent data leakage. In both validation frameworks, an internal validation subset was used to optimize the classification threshold based on the F2-score prior to final model training. This unified experimental design ensures a consistent and comparable evaluation under both Stratified and spatially explicit

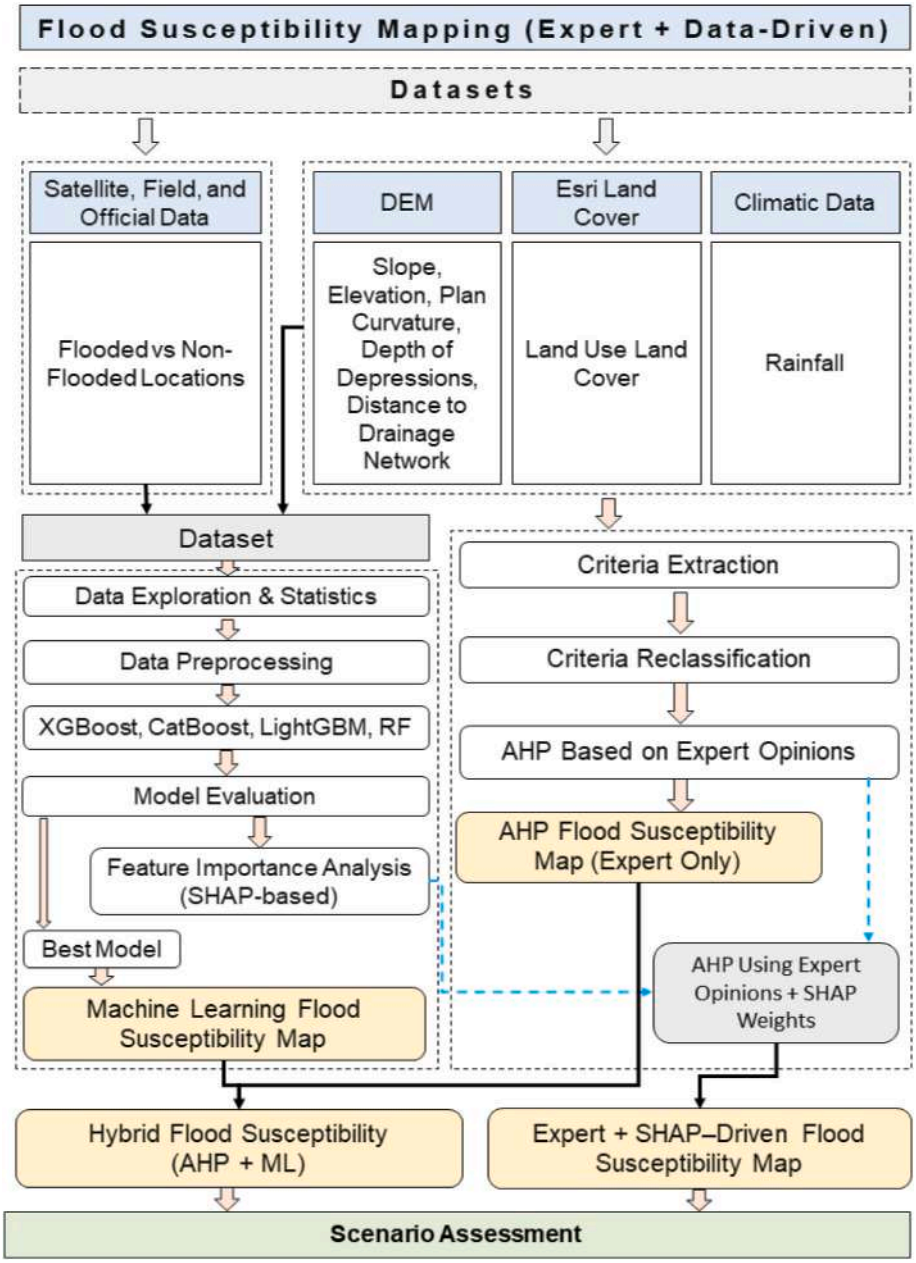


Fig. 2. Workflow of the integrated interpretable ML and MCDA framework for flood susceptibility mapping.

conditions, while providing a robust assessment of model generalization performance. For each fold, standard performance metrics including accuracy, ROC–AUC, precision, recall, and F1-score were computed, while an optimal classification threshold was derived within each fold using an internal validation split based on the F2-score. This procedure was consistently applied across both stratified and spatial cross-validation frameworks to ensure a fair and comparable evaluation. Final results were summarized by averaging fold-wise metrics, and an overall confusion matrix was obtained by aggregating predictions across folds, thereby reducing potential bias associated with a single train–test split (Grekousis, 2025; Teodorescu and Obreja Brașoveanu, 2025). All fold-level outputs, optimized thresholds, hyperparameter configurations, and aggregated confusion matrices are reported in Appendix B (Tables B1–B12 and Equation (B1)), whereas mean performance metrics are presented in the Results section for model comparison.

We used SHAP to quantify feature contributions within the predictive models. For each spatial cross-validation fold, SHAP values were computed using a TreeExplainer and aggregated in absolute form to reflect feature importance independent of direction. Fold-wise results were then averaged across all spatial folds to obtain global importance scores, which were normalized to a 0–1 scale. For LULC, the SHAP values of the one-hot encoded categories were summed into a single composite feature (LULC_Total) to avoid fragmentation of importance. These SHAP values, derived within the spatial cross-validation framework, were subsequently used as the basis for interpretation and MCDA integration, as they better capture spatially independent model behaviour and mitigate bias associated with spatial autocorrelation. Building on the same validation framework, the best-performing model was selected for spatial prediction and applied to the entire study area using the seven conditioning factors as raster inputs. The resulting pixel-wise outputs were used to generate a continuous flood susceptibility surface, which was subsequently classified into susceptibility levels for mapping and comparative analysis, ensuring full consistency between model interpretation, evaluation, and spatial prediction.

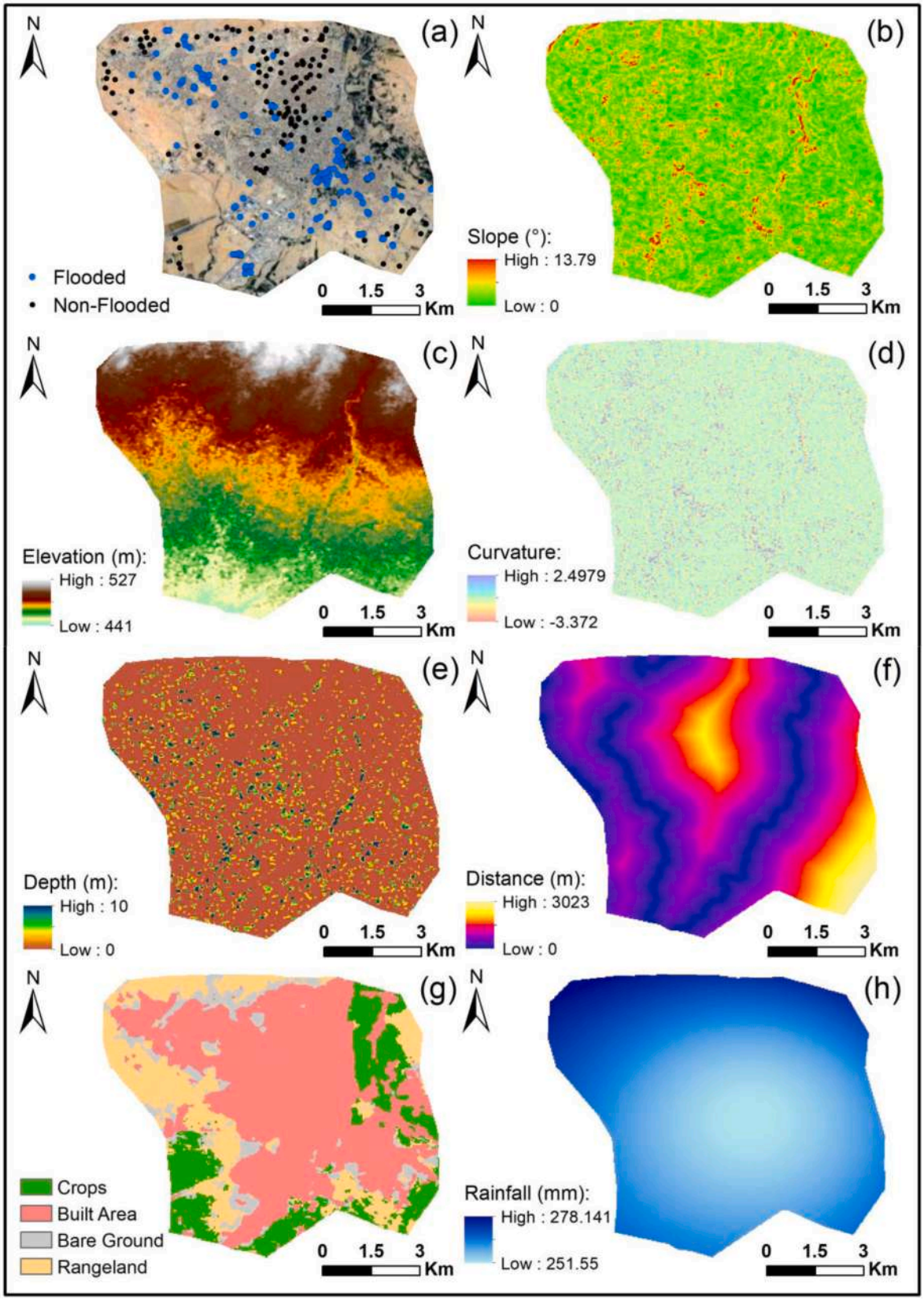


Fig. 3. Spatial distribution of conditioning factors and flood occurrence points in the study area: (a) flooded and non-flooded points, (b) Slope, (c) Elevation, (d) Plan Curvature, (e) Depth of Depressions, (f) Distance to the Drainage Network, (g) Land use/land cover (LULC), and (h) Precipitation.

2.5. AHP-based flood susceptibility mapping

2.5.1. Reclassification of conditioning factors

After deriving the conditioning criteria in the Data Preparation stage, all criteria were reclassified to a unified 1–5 scale, representing susceptibility from very low (1) to very high (5). Reclassification was performed in ArcGIS Pro based on prior knowledge of the study area and reference to previous studies. The Natural Breaks (Jenks) method was applied to some criteria, as it is commonly used in flood susceptibility studies (Table 1 summarizes the reclassification thresholds for all criteria).

2.5.2. Expert Pairwise Comparisons

An online survey was designed and distributed to experts with knowledge of the study area and a general understanding of MCDA. Seven experts responded, bringing expertise from disciplines such as urban planning and urban natural hazards. Each expert completed a pairwise comparison matrix for the selected criteria using the Saaty scale (1–9) (Saaty, 1987). To ensure consistency within each matrix and validate the reliability of the assigned weights, we calculated the Consistency Ratio (CR). The Consistency Index (CI) and Random Consistency Index (RI) were determined prior to computing the CR, with RI set to 1.32 for seven criteria. All CR values were below the threshold of 0.1, confirming that expert judgments were consistent. Detailed formulas for CI and CR are provided in Appendix C (Table C1 and Equation (C1)).

We then computed the geometric mean of the seven matrices to produce a final aggregated matrix, a method widely used for combining expert judgments in AHP (Krejčí and Stoklasa, 2018; Ossadnik et al., 2016). The CI, CR, and RI were also calculated for the aggregated matrix to confirm the reliability of the combined weights. The seven individual matrices are provided in Appendix C (Tables C2–C8), while Table 2 presents the aggregated pairwise comparison matrix, the resulting criterion weights, and the overall consistency ratio.

Table 1
Reclassification of conditioning factors and flood susceptibility levels.

Criteria	Classes	Susceptibility
Slope (°)	0 - 2.50	very high
	2.51 - 5	high
	5.01 - 8	Moderate
	8.01 - 11	low
	11.01 - 13.79	very low
Elevation (m)	441 - 460	very high
	460.01 - 480	high
	480.01 - 490	Moderate
	490.01 - 500	low
	500.01 - 527	very low
Plan Curvature	−3.37 - 0	Moderate
	0	very high
	0 - 2.50	low
Depth of Depressions (m)	0 - 1	very low
	1.01 - 2	low
	2.01 - 5	Moderate
	5.01 - 8	high
	8.01 - 10	very high
Distance to the Drainage Network (m)	0 - 300	very high
	300.01 - 600	high
	600.01 - 900	Moderate
	900.01 - 1500	low
	1500.01 - 3023	very low
Land use/land cover	Rangeland	very low
	Crops	low
	Built Area	very high
	Bare Ground	Moderate
Precipitation (mm)	251.55 - 255.83	very low
	255.84 - 259.92	low
	259.93 - 264.56	Moderate
	264.57 - 269.92	high
	269.923 - 278.14	very high

2.5.3. Derivation of the AHP-based flood susceptibility map

Using a GIS-based weighted overlay approach, the expert-derived weights obtained from the Expert Pairwise Comparisons step were combined with the reclassified criteria from the Reclassification of Conditioning Criteria step. Flood susceptibility was calculated according to the following equation:

$$\mathrm{FSM} = \sum_{i=1}^{n} (\mathrm{Criteria_i} \times \mathrm{Weight_i}) \quad (1)$$

Where:

- *Criteria _i:* The reclassified value of factor i
- *Weight_i:* The expert-derived weight for Criteria i
- *n:* The total number of Criteria used

To facilitate interpretation and subsequent comparison, the final flood susceptibility map was reclassified into five classes ranging from very low to very high susceptibility using the Natural Breaks (Jenks) method, thereby enabling direct comparison with the results of the subsequently proposed scenarios.

2.6. Integration scenarios for flood susceptibility mapping

2.6.1. Scenario I: map-level integration of machine learning and AHP results

After evaluating the four ML models, CatBoost was identified as the optimal model based on its superior predictive accuracy and robustness. Using the same seven conditioning factors, the trained CatBoost model was applied to the entire study area to generate a data-driven flood susceptibility map. The expert-driven flood susceptibility map obtained in the previous AHP step served as the reference for integration. Prior to combining the maps, the machine learning–based map was rescaled to a common range to ensure compatibility with the AHP-derived map. The two maps were then integrated at the raster level using a weighted overlay approach implemented in a GIS environment (Raster Calculator). Considering the limited size of the training dataset and the complementary nature of both approaches, a weight of 0.6 was assigned to the AHP-based map as the knowledge-driven component, while the ML map contributed a weight of 0.4 to incorporate data-driven evidence and enhance spatial generalization. Finally, the integrated flood susceptibility map was reclassified into five discrete classes, ranging from very low to very high susceptibility, using the Natural Breaks (Jenks) method, enabling direct comparison with the subsequently proposed scenario.

2.6.2. Scenario II: weight-level integration of SHAP and expert judgments within an AHP framework

In this scenario, integration was performed at the weight level rather than the map level. Mean absolute SHAP values were extracted from all four ML models, rather than relying solely on the best-performing model. These values were into unitless weights (0–1 scale) (see Table 3).

The normalized SHAP weights were then aggregated across the four models using a simple arithmetic mean to derive a consensus-based importance estimate for each conditioning factor. This aggregation strategy preserves the quantitative information derived from the models while avoiding additional subjective transformation and ensures equal contribution of all models, justified by their comparable predictive performance. Finally, the SHAP-based consensus weights and the AHP-derived weights were integrated using a weighted fusion approach, resulting in a final composite weight that reflects both data-driven model explanations and expert knowledge. The final integrated weight for each factor is expressed as follows:

$$\boldsymbol{W}_{combined} = \boldsymbol{W}_{expert}^{0.6} \times \boldsymbol{W}_{SHAP}^{0.4} \quad (2)$$

The parameters of the previous equations (0.6) and (0.4) were

Table 2
Aggregated pairwise comparison matrix based on expert judgments, criterion weights, and consistency ratio (CR).

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C1	1	1.47	1.74	1.84	0.58	1.29	1.92	**0.18**	**2**
C2	0.68	1	1.04	1.15	0.47	0.77	1.10	**0.12**	**4**
C3	0.58	0.96	1	1.10	0.51	0.64	0.94	**0.11**	**6**
C4	0.54	0.87	0.91	1	0.48	0.58	0.85	**0.10**	**7**
C5	1.71	2.12	1.96	2.10	1	1.64	2.14	**0.24**	**1**
C6	0.77	1.29	1.57	1.74	0.61	1	1.43	**0.15**	**3**
C7	0.52	0.91	1.06	1.17	0.47	0.70	1	**0.11**	**5**
CI	0.004								
RI	1.32								
CR	**0.003**								

C1: Slope, **C2:** Elevation, **C3:** Plan Curvature, **C4:** Depth of Depressions, **C5:** Distance to the Drainage Network, **C6:** Land use/land cover, **C7:** Precipitation.

Table 3
Mean absolute SHAP values and normalized importance weights (0–1 scale) for each feature across the four machine learning models.

	XGBoost		CatBoost		LightGBM		RF	
	Mean_SHAP	(%)	Mean_SHAP	(%)	Mean_SHAP	(%)	Mean_SHAP	(%)
F1	0.54	0.11	0.54	0.08	0.87	0.12	0.03	0.06
F2	1.2	0.25	1.99	0.29	1.79	0.25	0.14	0.28
F3	0.25	0.05	0.25	0.04	0.43	0.06	0.02	0.04
F4	0.26	0.05	0.51	0.08	0.43	0.06	0.04	0.08
F5	1.36	0.28	1.81	0.27	2.01	0.28	0.13	0.26
F6	0.79	0.16	0.86	0.13	0.86	0.12	0.07	0.14
F7	0.49	0.10	0.84	0.12	0.67	0.09	0.07	0.14

F1: Slope, **F2:** Elevation, **F3:** Plan Curvature, **F4:** Depth of Depressions, **F5:** Distance to the Drainage Network, **F6:** Land use/land cover, **F7:** Precipitation.

defined to assign a slight preference to expert knowledge while preserving the explanatory contribution of SHAP-derived weights from ML models as complementary quantitative evidence rather than an independent decision rule, which in turn allows direct comparability with the previous map-level integration scenario. Subsequently, the fused weights were normalized by dividing each value by the total sum to ensure consistency with the weighted overlay analysis framework (see Table 4).

The resulting combined weights were subsequently applied using a GIS-based weighted overlay approach similar to that employed in the AHP map, but using the final weights (Equation (3)), and integrating them with the reclassified conditioning criteria to produce a final flood susceptibility map. The map was then reclassified into five discrete classes, ranging from very low to very high susceptibility, facilitating direct comparison with Scenario I.

$$FSM_{combined} = \sum_{i=1}^{n} (\text{Criteria}_i \times \text{W_combined}) \quad (3)$$

2.7. Sensitivity Analysis of Hybrid Integration Scenarios

Sensitivity analysis was conducted to evaluate the stability of the final flood susceptibility results under variations in the weights assigned to the data sources used in Scenario I and Scenario II. The reference weights (0.6 for the knowledge-driven component and 0.4 for the ML component) were adopted, and four alternative weighting configurations were considered: the inverse weighting case (0.4 for the knowledge-driven component and 0.6 for the ML component), an equal weighting case (0.5/0.5), and two extreme weighting cases favouring either the knowledge-driven component (0.7/0.3) or the ML component (0.3/0.7).

These weighting configurations were applied to both scenarios, and their effects on the final flood susceptibility results were evaluated. The assessment was based on the relative change in susceptibility class areas using the percentage change formulation (Equation (4)).

$$\%\Delta = \frac{\mathbf{Ascenario} - \mathbf{Areference}}{\mathbf{Areference}} \times \mathbf{100} \quad (4)$$

In addition, spatial stability was assessed using the Kappa coefficient, allowing for the evaluation of spatial consistency across the different weighting configurations.

Table 4
Expert-derived weights, SHAP-based weights, fused weights, and final normalized weights for weighted overlay analysis.

Criteria	AHP weight	SHAP weight (mean of 4 models)	Fused weight	Final weight
Slope	0.18	0.09	0.14	0.14
Elevation	0.12	0.27	0.16	0.17
Plan Curvature	0.11	0.05	0.08	0.08
Depth of Depressions	0.10	0.07	0.08	0.09
Distance to the Drainage Networ	0.24	0.27	0.25	0.26
Land use/land cover	0.15	0.14	0.15	0.15
Precipitation	0.11	0.12	0.11	0.11

2.8. Assessment scenario

Given the scarcity of detailed reference data, we adopted a relative and comparative evaluation strategy instead of strict validation. We consulted Some previous studies conducted for the same city (Hadji and Felloussia, 2023; Loumi, 2022; Tarek et al., 2023), which relied on governmental records and field investigations to identify historically flood-prone areas and low flood-risk zones. Reference information was extracted exclusively for six selected areas, including three historically exposed zones and three low-risk zones. Within these areas, reference points were generated with a uniform spacing of 100 m then a 100 m buffer was applied around all training samples, and only reference points located outside this buffer were retained to ensure spatial independence between training and evaluation datasets for the evaluation procedure. This approach was adopted to reduce spatial autocorrelation and facilitate ROC curve computation. This metric was selected due to its threshold-independent nature and suitability under limited reference data conditions, with AUC–ROC being computed using the continuous susceptibility outputs prior to reclassification into five classes (from very

low to very high susceptibility), in order to preserve the full spatial variability of the original values without information loss caused by classification. Although this framework does not constitute a strict validation, we consider that it provides a reasonable and defensible basis for comparing the relative performance and interpretability of stand-alone and integrated approaches under realistic data constraints (see Appendix D, Figs. D1–D2 and Table D1 for the selected reference areas, their spatial representation, and recorded flood recurrence).

3. Results

3.1. Machine learning models performance evaluation and application

3.1.1. Results under stratified cross-validation

Table 5 summarizes the mean performance across the three cross-validation folds of the ML models used in this study. Overall, all models demonstrated strong and reliable classification capability in distinguishing flooded from non-flooded samples, with CatBoost and LightGBM exhibiting the best overall performance in terms of accuracy and stability. LightGBM achieved the highest mean overall accuracy (85.87% ± 1.23), followed closely by CatBoost (85.19% ± 4.60), indicating more consistent performance across folds. In contrast, XGBoost yielded the lowest mean accuracy (80.71% ± 4.26), while Random Forest reached 81.73% ± 1.73, reflecting moderate yet stable predictive performance.

At the class level, LightGBM and CatBoost exhibited a well-balanced trade-off between precision and recall, as reflected in their F1-scores, for both flooded and non-flooded classes. Their class-wise metrics remained within a relatively high range, spanning from 79.29% mean recall for the non-flooded class in LightGBM to 92.00% mean recall for the flooded class in the same model, with all precision, recall, and F1-score values for both models falling within these bounds. This pattern indicates that both models maintained strong discriminative ability across classes, with a slight advantage in favor of LightGBM in terms of overall robustness and fold-to-fold stability.

Table 5
Stratified cross validated performance metrics of machine learning models for flooded and non flooded classes.

XGBoost				
		Mean Precision (%)	Mean Recall (%)	Mean F1-score (%)
Non- Flooded		87.43	70.76	77.69
Flooded		77.15	90.00	82.86
Accuracy (%)	Mean Accuracy	80.71		
	Standard Deviation	±04.26		
ROC-AUC (%)	Mean ROC-AUC	92.90		
	Standard Deviation	±03.22		
CatBoost				
		Mean Precision (%)	Mean Recall (%)	Mean F1-score (%)
Non- Flooded		88.20	80.77	83.87
Flooded		83.76	89.33	86.18
Accuracy (%)	Mean Accuracy	85.19		
	Standard Deviation	±04.60		
ROC-AUC (%)	Mean ROC-AUC	94.51		
	Standard Deviation	±02.21		
LightGBM				
		Mean Precision (%)	Mean Recall (%)	Mean F1-score (%)
Non- Flooded		90.423	79.29	84.39
Flooded		82.73	92	87.06
Accuracy (%)	Mean Accuracy	85.87		
	Standard Deviation	±01.23		
ROC-AUC (%)	Mean ROC-AUC	94.63		
	Standard Deviation	±02.12		
RF				
	Mean Precision (%)		Mean Recall (%)	Mean F1-score (%)
Non- Flooded	91.58		69.25	78.29
Flooded	76.93		93.33	84.09
Accuracy (%)	81.73 ±01.73			
ROC-AUC (%)	94.52 ±02.96			

3.1.2. Results under spatial cross-validation

Table 6 summarizes the mean performance of the four ML models evaluated in this study using three-fold spatial cross-validation. Overall, all models exhibited a moderate and relatively consistent classification performance in distinguishing between flooded and non-flooded samples. The results indicate a general reduction in predictive accuracy under spatial constraints, with CatBoost achieving the best balance between predictive accuracy and stability across folds (73.30% ± 7.77), while Random Forest showed comparatively lower accuracy and higher variability (71.76% ± 13.76), reflecting less stable performance under spatially structured data. LightGBM and XGBoost exhibited similar performance levels, with accuracies of 72.66% ± 10.08 and 72.38% ± 8.16, respectively.

In terms of discriminative ability, the ROC-AUC results confirmed a generally good separation capability between flooded and non-flooded samples. CatBoost achieved the highest mean ROC-AUC (78.50% ± 11.46), followed by Random Forest (77.81% ± 13.21), while LightGBM and XGBoost recorded comparable values (76.54% ± 11.24 and 76.04% ± 12.24, respectively). These findings indicate that all models retained a satisfactory capacity to discriminate between the two classes under spatial validation.

At the class level, performance remained acceptable across all models, with a noticeable imbalance between flooded and non-flooded samples. Flooded samples consistently achieved higher recall values across models, reaching up to approximately 79% in XGBoost, whereas lower recall values were observed for the non-flooded class, particularly in XGBoost, indicating reduced sensitivity for this category. In contrast, CatBoost and LightGBM demonstrated relatively more balanced precision–recall behavior compared to the other models. Overall, these results confirm that all models retained satisfactory predictive capability under spatial cross-validation.

The aggregated confusion matrices, compiled separately for stratified and spatial cross-validation for each model by combining the results across their respective folds, provide a detailed assessment of each model's ability to correctly classify flood-prone and non-flood-prone instances. The complete confusion matrices are presented in Appendix B (Tables B11 and B12).

3.1.3. Flood susceptibility mapping using the selected machine learning model under spatial validation

Based on spatial cross-validation results, CatBoost outperformed the other models and showed greater stability. Despite a decline in accuracy compared to stratified validation, it retained consistent classification performance. Therefore, it was selected for final spatial prediction and flood susceptibility mapping (Fig. 4), which revealed a pronounced dominance of the two extreme susceptibility classes. A large portion of the study area is classified within the highest susceptibility class,

Table 6
Spatial cross-validated performance metrics of machine learning models for flooded and non-flooded classes.

XGBoost				
		Mean Precision (%)	Mean Recall (%)	Mean F1-score (%)
Non- Flooded		77.87	62.84	69.21
Flooded		68.05	79.02	73.10
Accuracy (%)	Mean Accuracy	72.38		
	Standard Deviation	±8.16		
ROC-AUC (%)	Mean ROC-AUC	76.04		
	Standard Deviation	±12.24		
CatBoost				
		Mean Precision (%)	Mean Recall (%)	Mean F1-score (%)
Non- Flooded		70.89	77.05	73.70
Flooded		74.83	68.40	71.29
Accuracy (%)	Mean Accuracy	73.30		
	Standard Deviation	±7.77		
ROC-AUC (%)	Mean ROC-AUC	78.50		
	Standard Deviation	±11.46		
LightGBM				
		Mean Precision (%)	Mean Recall (%)	Mean F1-score (%)
Non- Flooded		78.44	68.49	71.78
Flooded		69.62	74.33	71.23
Accuracy (%)	Mean Accuracy	72.66		
	Standard Deviation	±10.08		
ROC-AUC (%)	Mean ROC-AUC	76.54		
	Standard Deviation	±11.24		
RF				
		Mean Precision (%)	Mean Recall (%)	Mean F1-score (%)
Non- Flooded		75.74	72.83	73.13
Flooded		69.27	69.81	68.29
Accuracy (%)		71.76		
		±13.76		
ROC-AUC (%)		77.81		
		±13.21		

particularly along the main Oued. Significant parts of the urban area, especially in the eastern and western sections, as well as the entire industrial zone, also fall into this class. In contrast, areas of very low susceptibility are mainly located in the northern and central parts of the urban area, with scattered low-susceptibility patches along the periphery. Only a very small fraction of the area is classified into the intermediate classes (high, moderate, and low susceptibility). These observations are confirmed by the statistics in Table 7, which show that each of the two extreme classes covers roughly 34% of the study area, whereas the intermediate classes collectively occupy less than one-third of the total area.

3.2. AHP-based flood susceptibility map

The flood susceptibility map derived from the AHP approach, based on the judgments of seven experts, reveals a spatial pattern dominated by the lower susceptibility categories. As shown in Fig. 5, the very low and low classes occupy the largest proportion of the study area. These categories are predominantly distributed in the eastern part of the area and extend from the northern to the central sectors. In these zones, they are spatially constrained between areas classified as high and very high susceptibility. In contrast, the high and very high categories represent the least extensive portions of the study area. Their spatial distribution is characterized by a clear concentration along both sides of the main Oued networks, forming localized zones of elevated susceptibility. The moderate category covers an intermediate area and generally appears as a transitional zone between lower and higher classes, although in some locations it exhibits a more dispersed spatial pattern.

The area statistics presented in Table 8 further support these spatial patterns. The low category covers the largest share of the study area (28.76%), followed by the moderate (24.72%) and very low (20.83%) categories. In comparison, the high and very high categories occupy smaller proportions, accounting for 18.18% and 7.51% of the total area, respectively.

3.3. Scenario I: flood susceptibility map based on integration of CatBoost and AHP

The CatBoost model results were integrated with expert-based AHP judgments, producing the flood susceptibility map shown in Fig. 6. The combined map displays a relatively balanced distribution across the five susceptibility classes, with a slight predominance of the lowest class. Areas classified as the highest and high susceptibility are generally located along the main Oued corridors. The zones of highest susceptibility are closest to the Oued, followed by high and then the transitional moderate susceptibility areas. The eastern part of the city is primarily assigned to the high susceptibility class, despite its relative distance from the Oued. Zones of the lowest susceptibility extend from the northern to the central parts of the urban area, including the eastern part of the city, while low susceptibility areas appear in the southwestern part of the city. These spatial patterns are quantitatively summarized in Table 9, confirming that each class occupies a significant portion of the study area and that no single class dominates the overall distribution.

3.4. Scenario II: flood susceptibility map based on integration of SHAP-derived and expert weights

The SHAP-derived weights from the four ML models were combined with expert-based AHP weights to produce the flood susceptibility map shown in Fig. 7. The map shows clear separation of the five susceptibility classes with minimal fragmentation. Very high and high susceptibility areas are mainly located along the main Oueds corridors, with very high zones closest to the Oued. The eastern part of the urban area is predominantly low, with some moderate pixels. Very low susceptibility zones occur in the eastern section of the study area, while moderate areas generally act as transitional zones between low and high susceptibility.

Table 10 summarizes the area distribution. The low, very low, and moderate classes collectively cover more than 70% of the study area, while high and very high classes occupy 19% and 9.5%, respectively. These statistics confirm the spatial patterns observed in Fig. 7 and show that all five classes are represented across the study area, with varying coverage reflecting their spatial distribution.

3.5. Sensitivity analysis outcomes and scenario comparison

To avoid overloading the main text with tabulated information, the detailed area statistics are provided in Appendix E (Tables E1 and E2), including the reference class areas and the corresponding percentage changes computed using the equation presented in Section 2.7 (Sensitivity Analysis of Hybrid Integration Scenarios). In the main text, only the summary figures of the percentage changes are presented (Fig. 8) to facilitate clearer interpretation. The complete set of susceptibility maps

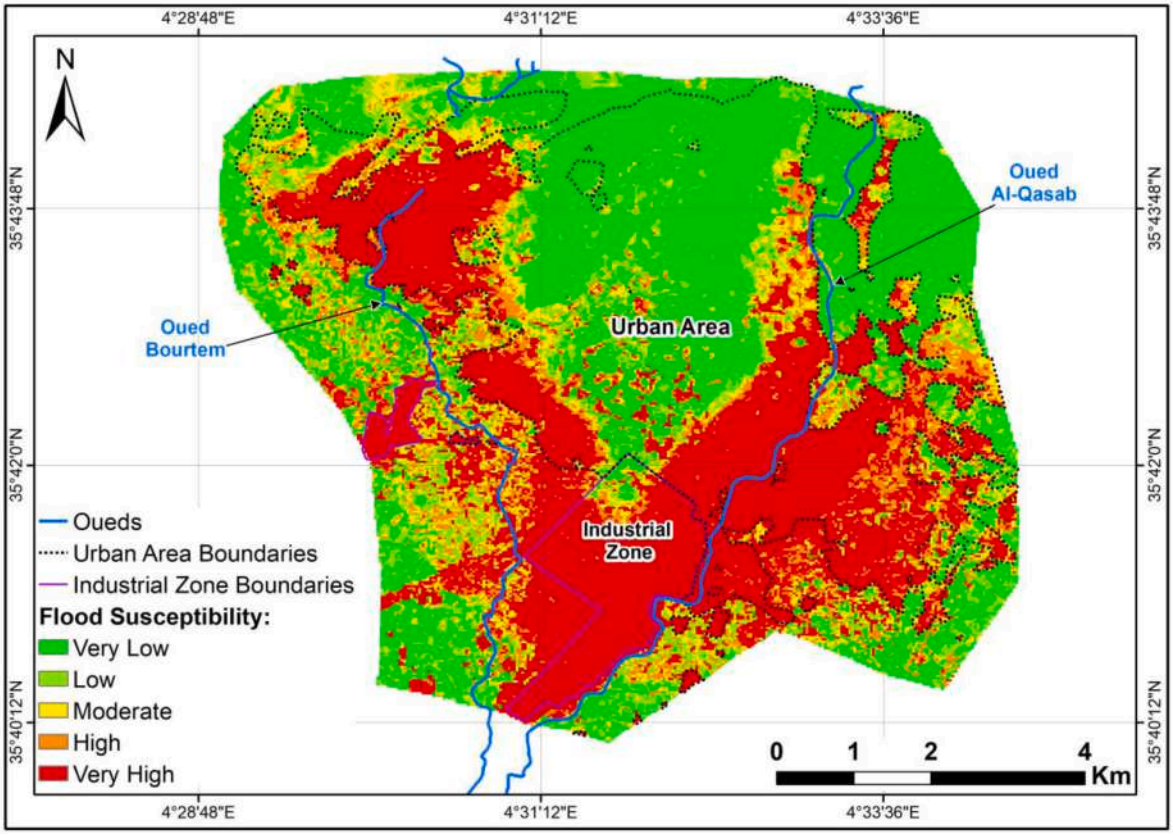


Fig. 4. Flood susceptibility map of the study area generated using the CatBoost model.

Table 7
Area distribution of flood susceptibility classes derived from the CatBoost-based flood susceptibility map.

Susceptibility	Area (square km)	(%)
Very Low	18.93	33.57
Low	7.40	13.13
Moderate	4.89	8.67
High	5.69	10.10
Very High	19.46	34.52

(eight maps in total), representing the two scenarios and their respective weighting configurations, is provided in the supplementary material to ensure clarity and maintain the conciseness of the manuscript.

The results indicate that Scenario I Fig. 8 (a) exhibits greater sensitivity to changes in the assigned weights than Scenario II Fig. 8 (b). In Scenario I, the percentage change ranges approximately from −34% to +57%, with the most pronounced variations occurring in the extreme susceptibility classes. In particular, the Very High class shows substantial increases when the weighting scheme shifts away from the baseline configuration of 0.6 for human expertise and 0.4 for the machine-learning model toward a higher contribution of the latter, reaching approximately +39% under the 0.4/0.6 configuration and up to +57% under 0.3/0.7. Similarly, the Very Low class also records notable increases of around +21% and +39%, respectively. In contrast, the intermediate classes (Low, Moderate, and High) consistently exhibit reductions in area, reaching approximately −34% across the different weighting configurations. Although the balanced 0.5/0.5 configuration produces comparatively moderate variations, the overall redistribution pattern remains evident, with a clear shift in area proportions toward the extreme classes as the weight balance moves toward the machine-learning component. By contrast, Scenario II demonstrates a considerably higher level of stability, with percentage changes remaining confined within a narrow interval of approximately −6% to +6% across all classes, indicating only minor adjustments in the spatial distribution of susceptibility classes in response to changes in the Weighting Configuration.

The spatial agreement statistics reported in Table 11 are consistent with the percentage change analysis and reveal clear differences between the two scenarios across all weighting configurations. Scenario II consistently yields higher agreement values than Scenario I. In Scenario II, Kappa values range from 0.87 to 0.96, with the highest value obtained under the 0.5/0.5 configuration (0.96) and the lowest under 0.3/0.7 (0.87). The corresponding Po values vary from 0.90 to 0.97, while Pe remains nearly constant at approximately 0.22. In contrast, Scenario I shows lower and more variable agreement, with Kappa ranging from 0.59 to 0.87, Po from 0.67 to 0.89, and Pe remaining relatively stable between 0.20 and 0.21. Across both scenarios, the 0.5/0.5 configuration produces the highest agreement, whereas the 0.3/0.7 configuration yields the lowest Kappa values.

3.6. Assessment of flood susceptibility mapping results

Fig. 9 presents the AUC–ROC curves for all models and integration scenarios. The standalone AHP model and the AHP + SHAP integration scenario achieved the highest discriminative performance, both recording an AUC of 0.964. The AHP + CatBoost integration followed closely with an AUC of 0.958, while standalone CatBoost yielded the lowest AUC of 0.859.

Table 12 presents the distribution of points for Historical Flood Locations (Point) and Low Flood Risk Locations (Point) across susceptibility classes derived from the outputs of the models and proposed integration scenarios used in this study, after reclassification into five classes ranging from Very Low to Very High. For the Historical Flood Locations (n = 73), the CatBoost model assigns the vast majority of points to the Very High class (70 points), while the AHP + CatBoost

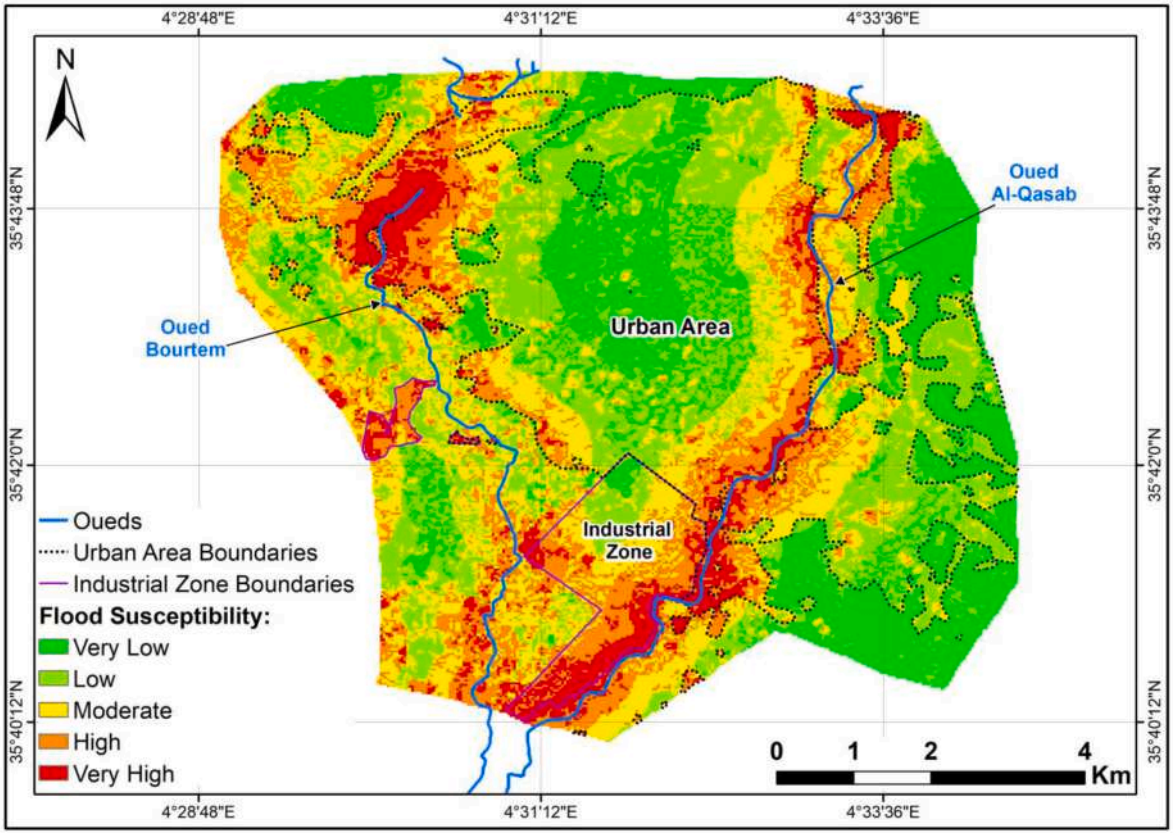


Fig. 5. Flood susceptibility map of the study area generated using the AHP method.

Table 8
Area distribution of flood susceptibility classes derived from the AHP-based susceptibility map.

Susceptibility	Area (square km)	(%)
Very Low	11.74	20.83
Low	16.21	28.76
Moderate	13.93	24.72
High	10.25	18.18
Very High	4.24	7.51

integration scenario shows a similar concentration in the same class (66 points). By comparison, AHP and the AHP + SHAP scenario distribute these points mainly between the High and Very High classes, with 28 and 38 points for AHP, and 28 and 37 points for AHP + SHAP, respectively, whereas the lower classes contain only a negligible number of points.

For the Low Flood Risk Locations (n = 139), AHP and the AHP + SHAP show a clear concentration in the lower susceptibility classes (Very Low, Low, and Moderate), with only one point assigned to the Very High class in both cases. Specifically, AHP records 31, 45, and 50 points in these classes, respectively, while AHP + SHAP shows a similar distribution with 24, 53, and 48 points. In contrast, the CatBoost model assigns a substantial portion of low-risk points to the Very High class (63 points), whereas the AHP + CatBoost presents a more dispersed distribution across classes, including 19 points in the same class.

4. Discussion

4.1. Model performance and feature contributions

All four ML models applied in this study (XGBoost, CatBoost, LightGBM, and Random Forest) demonstrated reliable classification performance in distinguishing flooded from non-flooded classes under standard cross-validation. However, their performance declined under spatial cross-validation, with overall accuracy decreasing by approximately 8% to 13% and ROC-AUC by roughly 16% to 18%. This reduction is expected, as spatial cross-validation represents a more stringent and realistic evaluation framework that reduces spatial dependence between training and testing samples. In contrast, standard or stratified cross-validation may yield overly optimistic estimates when spatial autocorrelation is present, as geographically proximate samples tend to share similar environmental characteristics, leading to inflated predictive performance. This issue was also observed in the present study, where strong spatial autocorrelation, as indicated by Moran's I, suggests that stratified partitioning may overestimate model performance. Similar findings have been reported in previous studies, which demonstrate that ignoring spatial structure can lead to biased and overoptimistic model evaluation (Marco et al., 2022; Mushagalusa et al., 2024). Moreover, spatial cross-validation may lead to lower performance estimates because it forces models to extrapolate to unseen spatial conditions and limits the availability of similar environmental patterns during training, thus providing a more conservative but realistic assessment of model generalization. Importantly, this reduction reflects more realistic generalization rather than weaker models. Accordingly, spatial cross-validation was adopted as the primary criterion for model selection in this study.

Among the four models, CatBoost achieved the highest overall accuracy and stability across evaluation metrics in the spatial cross-validation framework, indicating comparatively superior predictive behavior. Comparable flood-related studies have also reported CatBoost as the best-performing model (Ajin et al., 2025; Gao et al., 2025; Phong et al., 2023), while other studies have highlighted better performance for LightGBM (Saber et al., 2022), XGBoost (Ma et al., 2021), or Random Forest (Abedi et al., 2022; Ren et al., 2024). Such variability underscores that model performance is highly dependent on dataset characteristics,

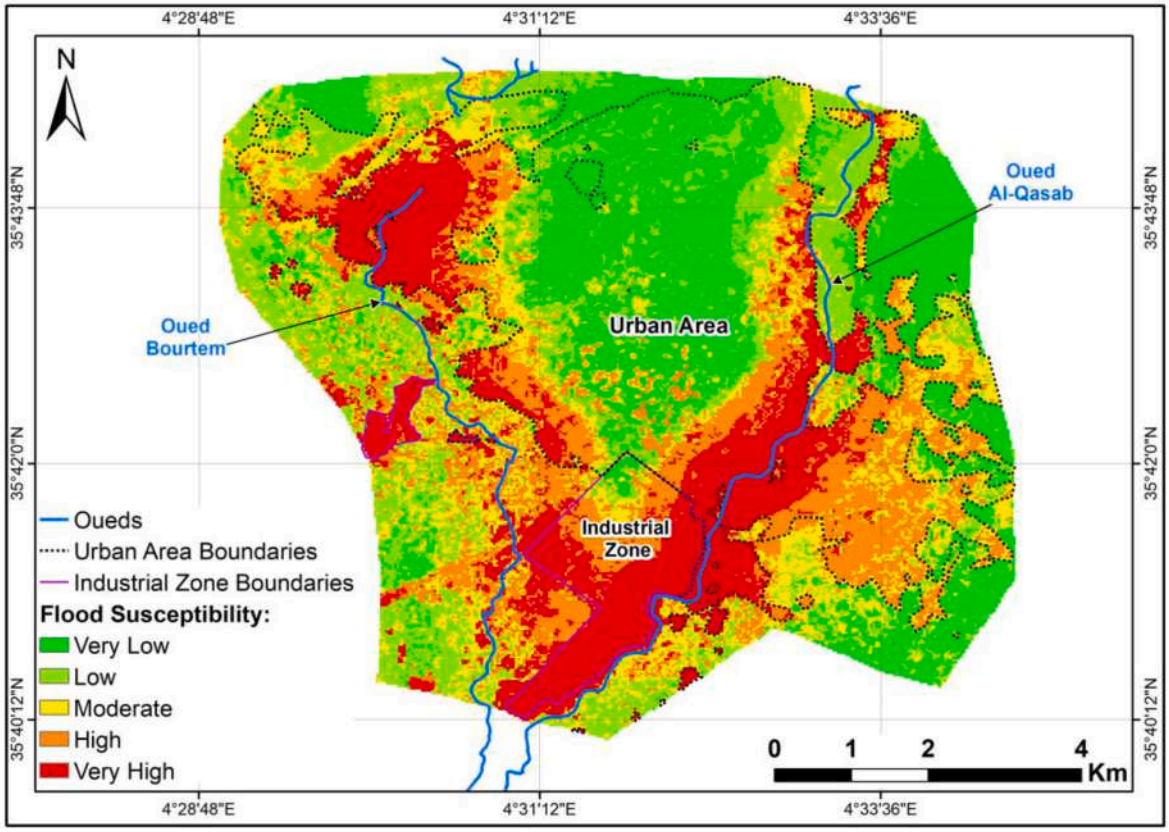


Fig. 6. Flood susceptibility map of the study area generated by integrating CatBoost and expert-based AHP results.

Table 9
Area distribution of flood susceptibility classes derived from the integrated CatBoost–AHP map (Scenario I).

Susceptibility	Area (square km)	(%)
Very Low	13.11	23.27
Low	12.79	22.70
Moderate	8.83	15.68
High	11.07	19.66
Very High	10.53	18.69

feature interactions, spatial context, and preprocessing strategies (Ajin et al., 2025). The consistently competitive performance observed across all tree-based models further confirms their general suitability for flood mapping applications, while emphasizing the necessity of case-specific validation and comparative assessment rather than reliance on a single modeling approach.

SHAP analysis revealed that Elevation, Distance to the Drainage Network, and Land use/land cover were the most influential factors driving model predictions. Their relative ranking remained consistent across models, although the contribution magnitudes varied slightly, reflecting model-specific sensitivity to feature interactions. Elevation and drainage proximity control water accumulation, flow paths, and inundation probability (Tsumita et al., 2025), while land use and land cover affect runoff, infiltration, and flood propagation (Bojer et al., 2025; Oo and Humphries, 2026). The alignment between the models' satisfactory performance and the dominance of these environmental factors indicates that the models are effectively capturing the primary controls of flooding. These findings are consistent with prior flood studies, which emphasize topography, drainage network proximity, land cover, and other interacting environmental factors as key determinants of flood behavior (Alam et al., 2025; Gao et al., 2025; Gul, 2025; Singha et al., 2025; Zhang et al., 2025). Overall, the results suggest that tree-based approaches reliably identify critical drivers of flood susceptibility while accounting for complex interactions among multiple environmental variables.

4.2. Convergence and divergence between SHAP-derived and expert-based weights

The comparison between SHAP-derived importance values and expert-based AHP weights reveals both notable consistency and meaningful divergence, providing the rationale for the proposed weight-level integration framework. Previous flood-related studies report substantial variability in both the selection and ranking of conditioning factors, reflecting the absence of a universally dominant driver of flood behavior. While some studies emphasize slope and river density (Vojtek and Vojteková, 2019) or precipitation and impervious surfaces (Pimenta et al., 2025), others identify elevation and distance to rivers as primary controls, alongside moderate contributions from drainage density, land use, vegetation, and rainfall (Ashfaq et al., 2025; Sharker et al., 2025). This variability highlights the context-dependent nature of flood susceptibility, shaped by geographic setting, data structure, and methodological choices. Consequently, core factors frequently reported in the literature (slope, elevation, land use/land cover, distance to rivers, and precipitation) were adopted in this study (Bouchikhi et al., 2025), with soil type excluded due to spatial homogeneity.

Within this study, expert-based weighting prioritized distance to drainage networks, slope, and land use/land cover, while assigning comparable influence to the remaining factors, a pattern consistent with studies emphasizing channel proximity and surface characteristics as dominant flood drivers (Pimenta et al., 2025; Vojtek and Vojteková, 2019). In contrast, SHAP-derived weights showed strong agreement with expert judgments for land use/land cover and distance to drainage networks, but indicated a higher relative importance of elevation compared to slope. This outcome aligns with previous investigations

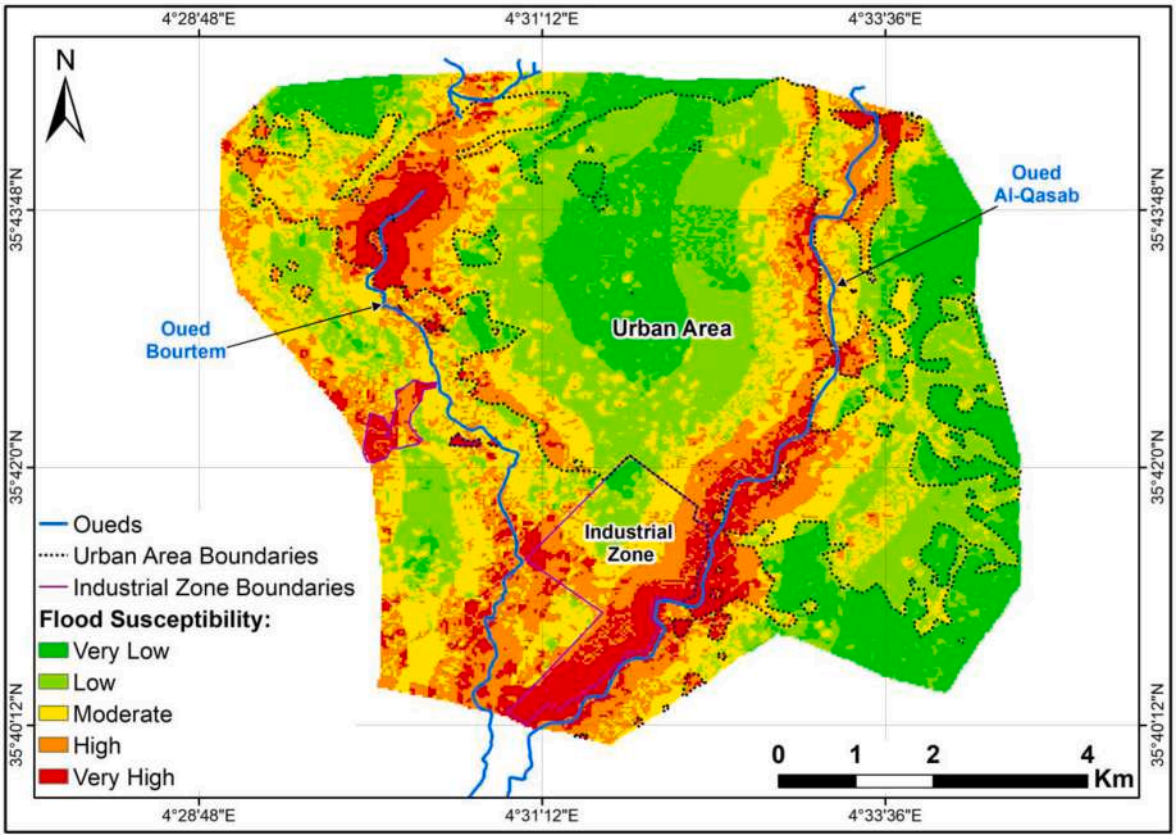


Fig. 7. Flood susceptibility map based on integration of SHAP-derived and expert AHP weights.

Table 10
Area distribution of flood susceptibility classes in the study area derived from Scenario II.

Susceptibility	Area (square km)	(%)
Very Low	10.09	17.90
Low	15.69	27.83
Moderate	14.46	25.65
High	10.78	19.13
Very High	5.35	9.49

that identified elevation as a dominant control on flood occurrence, particularly in low-relief or gently sloping environments (Ashfaq et al., 2025; Sharker et al., 2025).

The divergence in ranking between elevation and slope likely reflects their intrinsic interdependence and the different ways expert reasoning and data-driven learning capture their combined effects. While experts tend to emphasize slope as a dynamic process controlling runoff generation, SHAP captures elevation as a proxy for multiple interacting mechanisms, including flow convergence, accumulation zones, and drainage efficiency. Importantly, SHAP-based rankings were highly consistent across all ML models, indicating robust data-driven agreement. Overall, the partial convergence and systematic divergence between expert-based and SHAP-derived weights demonstrate that neither approach alone fully represents flood susceptibility drivers, thereby supporting the proposed weight-level integration as a balanced strategy that reduces expert bias while preserving physical interpretability.

4.3. Comparative analysis of map-level and weight-level integration strategies in flood susceptibility mapping

Several studies have shown that integrating ML outputs with multi-criteria decision analysis approaches can improve the reliability of flood susceptibility and flood risk mapping. For example, Adeyemi et al. (2024) reported superior performance for an integrated SVM–AHP model compared to individual approaches, while Nachappa et al. (2020) found that although ML models slightly outperformed MCDA models individually, the highest accuracy was achieved through model integration. Similar conclusions were drawn by other studies emphasizing the effectiveness of combining data-driven and decision-based methods for flood assessment (Bui et al., 2023; Liu et al., 2025; Shikhteymour et al., 2023). However, most existing studies focus on map-level integration, where independently generated outputs are combined after model execution, without modifying the internal weighting logic. In contrast, the present study prioritizes controlling model behavior, enhancing transparency, and reducing methodological bias by explicitly translating SHAP-derived feature importance values into normalized, transparent, and traceable weights within an AHP-based framework, thereby ensuring that each factor's contribution to the final map remains explicitly defined, interpretable, and directly justifiable, rather than solely maximizing numerical accuracy.

The results are consistent with previous studies showing that integrating ML outputs with expert knowledge improves model performance compared to standalone approaches, as reflected in the AUC values. However, map-level integration appears highly sensitive to weighting schemes, where small changes in the relative contribution of each source can lead to noticeable shifts in the final outputs, as confirmed by sensitivity and agreement analyses. This sensitivity likely arises because post hoc combination directly propagates discrepancies between input maps, particularly when one model exhibits a skewed or extreme class distribution, as observed for CatBoost. This behavior is likely influenced by the binary training structure of the ML model, which promotes sharp class separation and may reduce spatial gradation (Pugliese Viloria et al., 2024). In contrast, weight-level integration demonstrated substantially greater stability, which can be attributed to incorporating data-driven information at the weighting stage rather than at the final

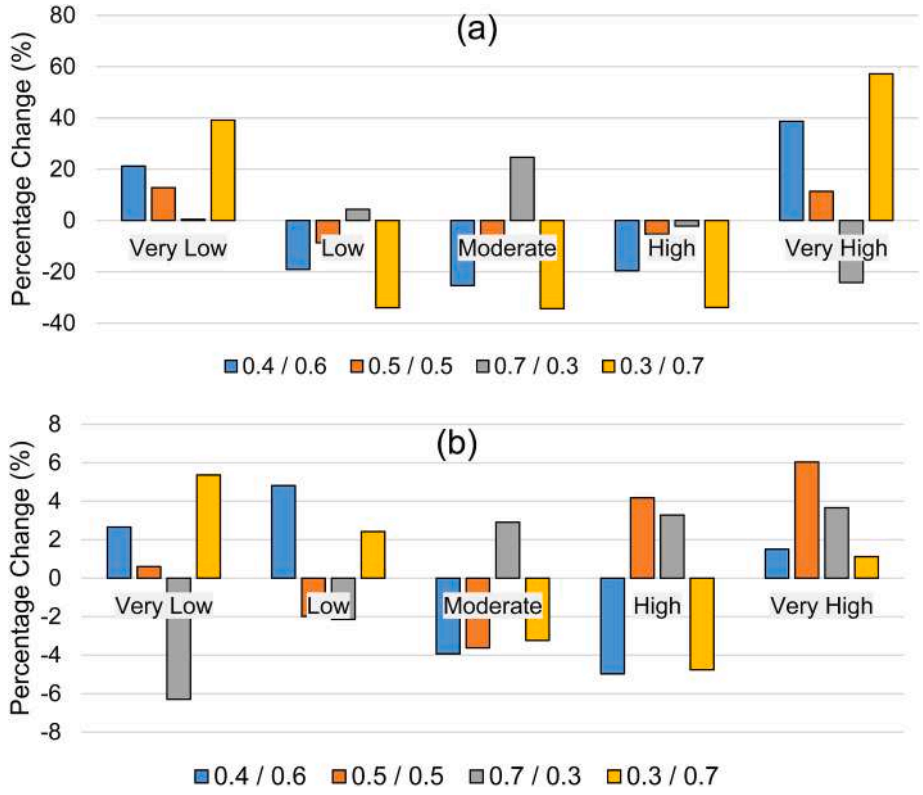


Fig. 8. Percentage change in susceptibility class areas under different weighting configurations.
(a) Scenario I: map-level fusion approach; (b) Scenario II: integration of SHAP-derived weights with AHP weights.
Note: The weighting configurations (e.g., 0.6/0.4, 0.5/0.5, and 0.4/0.6) indicate the relative contribution of the AHP-based expert weights (first value) and the machine learning-derived component (second value).

Table 11
Spatial agreement (**P_o**, **P_e**, **Kappa**) between reference and weight-modified flood susceptibility maps.

Weighting Configuration	Scenario I			Scenario II		
	P_o	P_e	Kappa	P_o	P_e	Kappa
0.4/0.6	0.78	0.21	0.72	0.93	0.22	0.90
0.5/0.5	0.89	0.20	0.87	0.97	0.22	0.96
0.7/0.3	0.84	0.20	0.80	0.96	0.22	0.95
0.3/0.7	0.68	0.20	0.59	0.90	0.22	0.87

Note: Values in the Weighting Configuration column represent the weights assigned to the knowledge-driven and machine learning components, respectively; P_o and P_e denote the observed and expected agreement, respectively, derived from pixel-based comparisons and expressed as proportions ranging from 0 to 1.

map stage, thereby reducing the impact of inter-model variability and producing more balanced and robust outputs.

4.4. Limitations and future research directions

This study employed a validation strategy combining stratified and spatial cross-validation in training the machine learning models to obtain realistic performance estimates. Spatially independent test points were used within six reference areas selected under data constraints and incorporated into the independent validation framework. Although their selection was justified in the study, no formal sensitivity analysis was conducted regarding the influence of area selection on the results, and its potential effect remains not fully quantified. The independent validation points were also limited in number and spatially confined to specific areas rather than being distributed across the entire study region, providing only a partial basis for accuracy assessment.

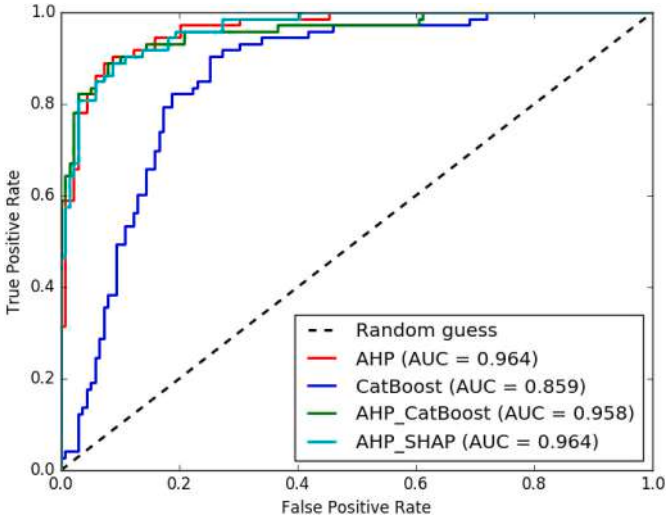


Fig. 9. AUC–ROC performance of individual and integrated flood susceptibility models.

Accordingly, the evaluation placed greater emphasis on internal consistency, relative model comparison, sensitivity analysis, agreement analysis, and spatial pattern interpretation rather than sole reliance on accuracy metrics. Future studies should therefore focus on increasing the spatial representativeness and density of validation samples, as well as systematically evaluating the effect of reference area selection on model performance and uncertainty. The analysis was also restricted to a single case study due to data availability constraints, which limits direct generalization and underscores the need for future studies to test the

Table 12
Distribution of reference points from historically flood-exposed and low flood-risk areas across flood susceptibility classes.

Historical Flood Locations (Point)				
	CatBoost	AHP	AHP + CatBoost	AHP + SHAP
Very Low	0	0	0	0
Low	2	1	2	0
Moderate	0	6	0	8
High	1	28	5	28
Very High	70	38	66	37
Low Flood Risk Locations (Point)				
Very Low	29	31	31	24
Low	24	45	29	53
Moderate	13	50	15	48
High	10	12	45	13
Very High	63	1	19	1

proposed framework in other regions and assess its transferability under different hydrogeomorphic conditions.

The AHP component was based on seven domain experts selected for their relevant expertise, and their judgments were internally consistent; nevertheless, a larger and more diverse panel would further strengthen representativeness. Interpretability was addressed structurally by explicitly translating SHAP-derived importance values into transparent, traceable weights and integrating them with expert-based weights within an MCDA framework, ensuring direct attribution of each factor in the final map. No formal or user-based interpretability assessment was conducted, so future work should focus on multi-source flood datasets, broader cross-regional validation, expanded expert elicitation, and formal evaluation of interpretability in decision-support contexts.

5. Conclusion

This study addressed a key methodological gap in flood susceptibility research related to transparency, interpretability, and decision relevance within urban climate risk assessment. To this end, an integrated GIS-based framework was developed that combines data-driven ML with expert-based MCDA, with the aim of supporting climate-informed environmental risk management. The framework compared conventional map-level integration with a weight-level integration strategy, in which model-specific SHAP-derived feature importance values were normalized, aggregated across the four ML models, and then fused with AHP-derived expert weights through source-level weighting.

The results indicate that all four tree-based ML models achieved reliable predictive performance, with CatBoost consistently showing the best overall accuracy and stability among them. However, performance generally declined under spatial cross-validation relative to standard stratified validation, highlighting the importance of using stricter spatially independent assessment. Map-level integration tended to preserve the differences between the contributing sources more directly, which can lead to susceptibility patterns dominated by stronger class contrasts and a limited representation of transitional zones. In contrast, the proposed weight-level integration improved structural transparency and spatial balance, while showing lower sensitivity to changes in source weights. Expert-derived and data-driven weights were recalibrated in a complementary manner, producing more balanced susceptibility maps, reducing false alarms, and better aligning the outputs with urban resilience planning and climate adaptation needs. This approach therefore shows promise as a practical tool for supporting climate-responsive spatial planning, risk communication, and policy-oriented flood management in semi-arid urban regions, although its transferability to areas with different hydro-climatic or ecological settings remains to be tested.

Overall, the findings highlight that effective flood susceptibility assessment should consider spatial validation, interpretability, and robustness alongside traditional accuracy metrics in the context of urban climate resilience. Future work could further strengthen the framework by incorporating more comprehensive flood inventories, additional conditioning factors, and cross-regional testing to improve its applicability and reliability across similar semi-arid urban environments.

Data and code

Public datasets used in this study (e.g., USGS DEM, ESRI Land Cover, and NASA POWER precipitation data) are described in the manuscript. The compiled flood and non-flood point datasets, as well as the model scripts and workflows, are available from the authors upon reasonable request, in accordance with the journal's data availability policy.

Use of AI tools in manuscript preparation

During the preparation of this work, the authors used ChatGPT to improve the writing style. After using this tool, the authors carefully reviewed and edited the content to ensure accuracy and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Sadiq Tahmi: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Xiaoping Rui:** Supervision, Writing – original draft, Writing – review & editing. **Soufiane Benzaoui:** Resources, Validation, Writing – original draft. **Meriem Khelali:** Investigation, Resources, Visualization, Writing – original draft, Writing – review & editing. **Housseyn Chebika:** Methodology, Resources, Writing – original draft, Writing – review & editing. **Imen Guechi:** Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial or non-financial interests that could have appeared to influence the work reported in this paper.

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Appendix. ASupplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envsoft.2026.107039.

Appendix A. Observed Flood Locations Supported by Landsat and Field Evidence *(Related to Section 2.3 Data Preparation)*

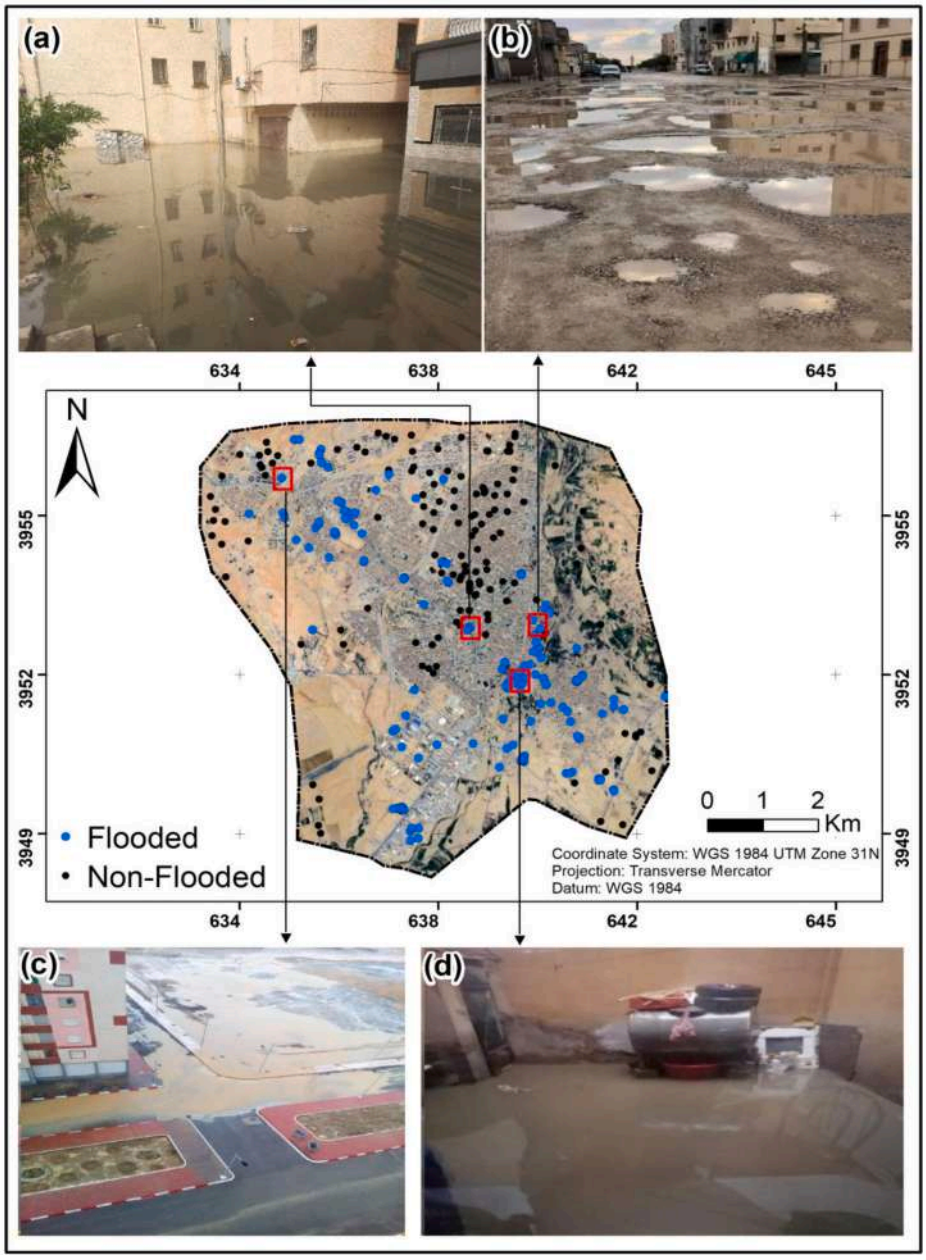


Fig. A1. Spatial distribution of selected flood-prone locations supported by visual evidence: (a) Field photograph of a repeatedly flooded site in the 8 May 1945 District (50 Duplex Housing), captured on 9 October 2022, with additional images confirming multi-year recurrence (Source: Chikhi tareq, personal communication, 2022); (b) El Kouche neighborhood (2021); (c) Nouvelle Cité AADL (2021); and (d) El Argoube neighborhood (2021) (Sources: Loumi, 2022).

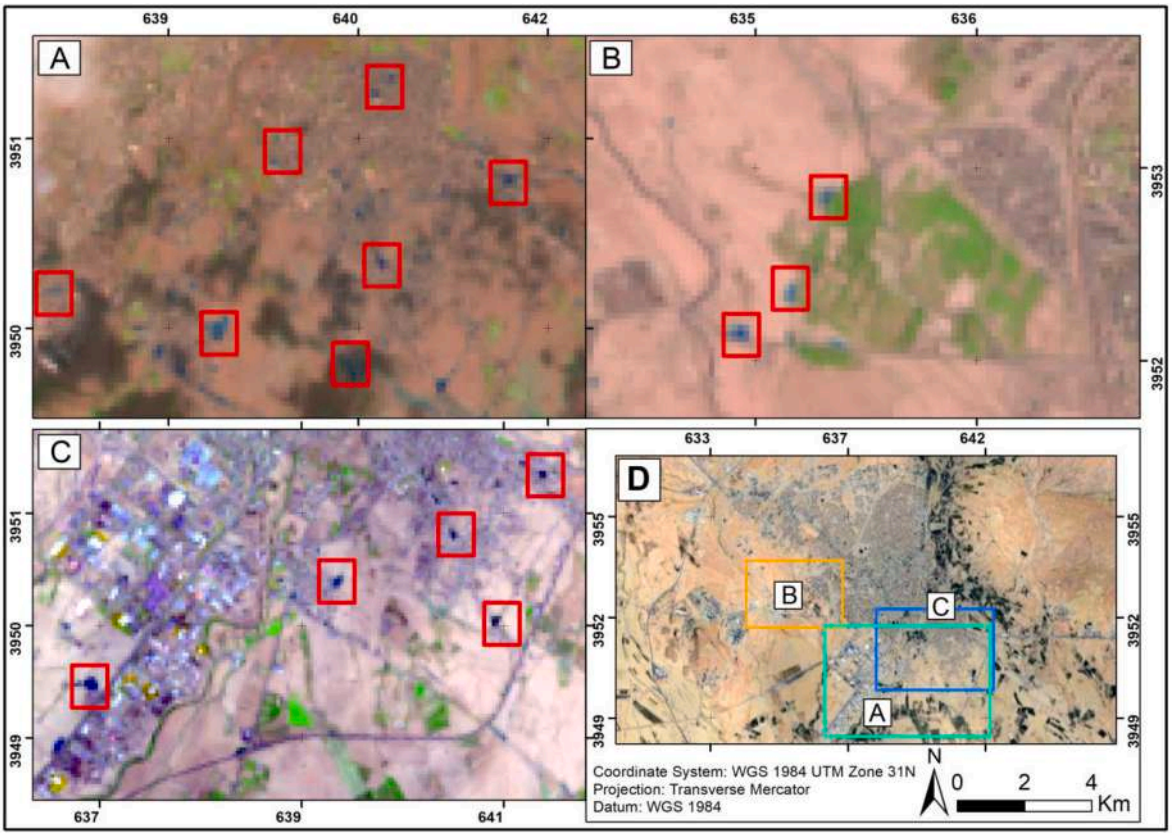


Fig. A2. Residual floodwater traces (water marks/deposits) identified from Landsat 8 imagery, indicating past flood events: (a) image acquired on 3 May 2021; (b) image acquired on 11 October 2013; (c) image acquired on 18 October 2021; and (d) spatial location of the analyzed scenes within the study area.

Appendix B. Spatial Distribution of Training Samples and Cross-Validation Fold Structure *(Related to Section 2.4 Machine Learning-Based Flood Susceptibility Mapping)*

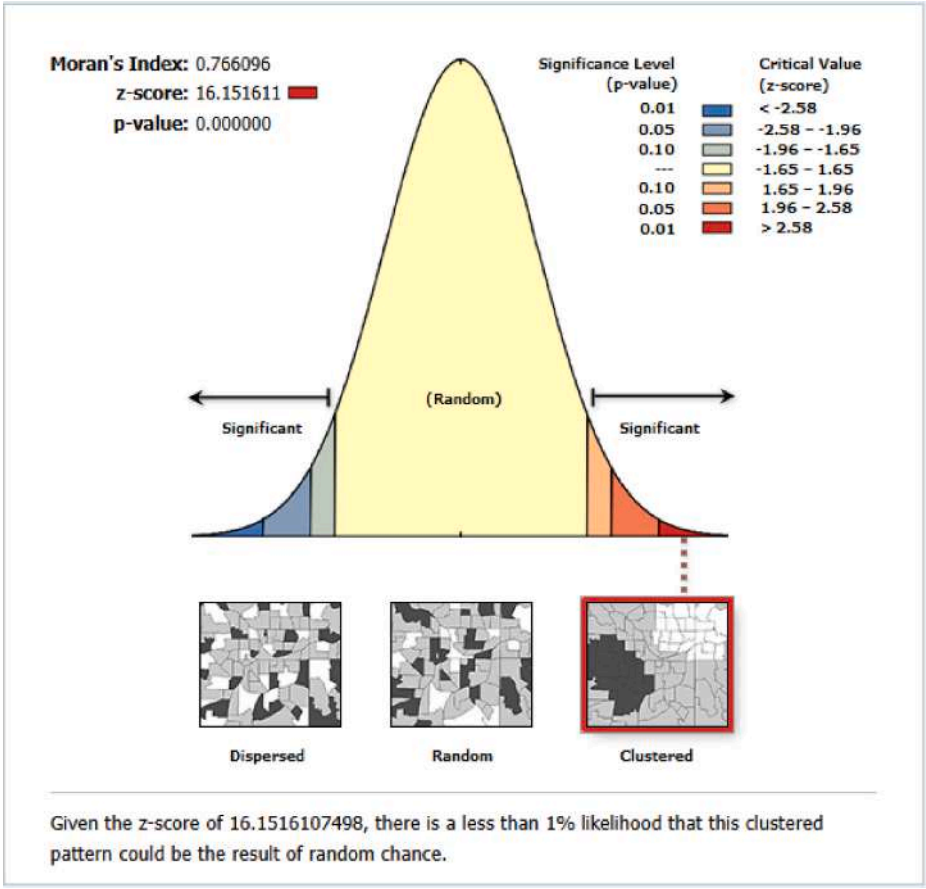


Fig. B1. Global Moran's I statistic for spatial autocorrelation of the binary flood occurrence dataset.

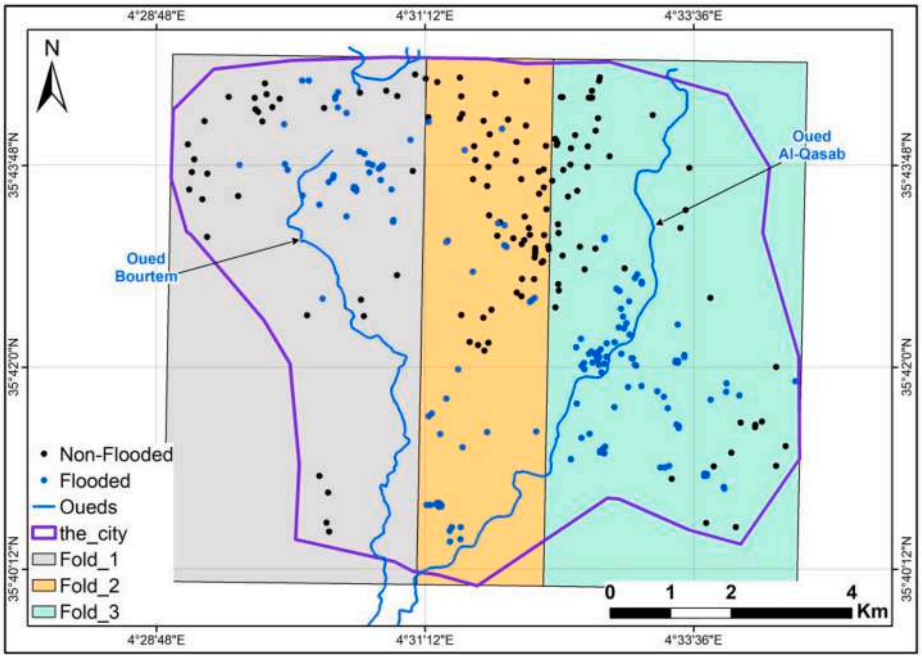


Fig. B2. Division of the study area into three spatial folds for cross-validation.

Table B1
Hyperparameter configurations of the XGBoost, CatBoost, LightGBM, and Random Forest models employed in this study

XGBoost		LightGBM	
Parameter	Value	Parameter	Value
n_estimators	450	n_estimators	500
learning_rate	0.05	learning_rate	0.08
max_depth	4	max_depth	5
min_child_weight	3	num_leaves	31
gamma	0.1	colsample_bytree	0.3
subsample	0.8	reg_alpha	1.2
colsample_bytree	0.8	reg_lambda	1.2
reg_alpha	0.5	objective	'binary'
reg_lambda	1.5	random_state	42
objective	'binary:logistic'	n_jobs	−1
eval_metric	'auc'	verbosity	−1
random_state	42		
n_jobs	−1		
CatBoost		**Random Forest**	
Parameter	Value	Parameter	Value
iterations	550	n_estimators	450
learning_rate	0.05	max_depth	9
depth	5	min_samples_split	2
l2_leaf_reg	4	min_samples_leaf	1
loss_function	'Logloss'	max_features	'sqrt'
eval_metric	'AUC'	random_state	42
random_seed	42	n_jobs	−1

Note: Hyperparameters for all models were first optimized using RandomizedSearchCV on the training dataset. The best-performing configurations were subsequently fixed and applied consistently across both spatial and random cross-validation to ensure an unbiased and fair performance comparison.

Equation (B1): Formulas for Precision, Recall, and F1-score

$$\text{Precision} = \frac{\textit{True Positives}}{\textit{True Positives} + \textit{False Positives}} \tag{1}$$

$$\text{Recall} = \frac{\textit{True Positives}}{\textit{True Positives} + \textit{False Negatives}} \tag{2}$$

$$\text{F1} - \text{score} = 2^{*}\ \frac{\textit{Precision}^{*}\textit{Recall}}{\textit{Precision} + \textit{Recall}} \tag{3}$$

Table B2
Optimal decision thresholds selected for each model across Stratified and Spatial cross-validation folds

Stratified cross-validation				
	XGBoost	CatBoost	LightGBM	Random Forest
Fold 1	0.20	0.15	0.23	0.32
Fold 2	0.27	0.43	0.20	0.39
Fold 3	0.29	0.34	0.20	0.22
Mean	**0.25**	**0.31**	**0.21**	**0.31**
Spatial cross-validation				
	XGBoost	CatBoost	LightGBM	Random Forest
Fold 1	0.20	0.25	0.48	0.41
Fold 2	0.39	0.23	0.20	0.45
Fold 3	0.37	0.28	0.21	0.45
Mean	**0.32**	**0.25**	**0.30**	**0.43**

Note: Threshold values were determined within each fold using an internal validation set by maximizing the F2-score. The reported values represent the optimal decision thresholds applied during model evaluation.

Table B3
Stratified cross-validation Results across All Folds for the XGBoost Model

Fold 1			
	Precision (%)	Recall (%)	f1-score (%)
Non-Flooded	90.00	57.00	70.00
Flooded	70.00	94.00	80.00
Accuracy (%)	76.29		
ROC-AUC (%)	93.32		
Fold 2			
Non-Flooded	80.00	77.00	78.00
Flooded	79.00	82.00	80.00
Accuracy (%)	79.38		
ROC-AUC (%)	88.77		
Fold 3			
Non-Flooded	92.00	78.00	85.00
Flooded	82.00	94.00	88.00
Accuracy (%)	86.46		
ROC-AUC (%)	96.61		

Table B4
Stratified cross-validation Results across All Folds for the CatBoost Model

Fold 1			
	Precision (%)	Recall (%)	f1-score (%)
Non-Flooded	91.00	68.00	78.00
Flooded	76.00	94.00	84.00
Accuracy (%)	81.44		
ROC-AUC (%)	93.87		
Fold 2			
Non-Flooded	80.00	85.00	82.00
Flooded	85.00	80.00	82.00
Accuracy (%)	82.47		
ROC-AUC (%)	92.17		
Fold 3			
Non-Flooded	93.00	89.00	91.00
Flooded	90.00	94.00	92.00
Accuracy (%)	91.67		
ROC-AUC (%)	97.48		

Table B5
Stratified cross-validation Results across All Folds for the LightGBM Model

Fold 1			
	Precision (%)	Recall (%)	f1-score (%)
Non-Flooded	92.00	74.00	82.00
Flooded	80.00	94.00	86.00
Accuracy (%)	84.54		
ROC-AUC (%)	94.38		
Fold 2			
Non-Flooded	87.00	83.00	85.00
Flooded	85.00	88.00	86.00
Accuracy (%)	85.57		
ROC-AUC (%)	92.17		
Fold 3			
Non-Flooded	93.00	80.00	86.00
Flooded	84.00	94.00	89.00
Accuracy (%)	87.50		
ROC-AUC (%)	97.35		

Table B6
Stratified cross-validation Results across All Folds for the Random Forest Model

Fold 1			
	Precision (%)	Recall (%)	f1-score (%)
Non-Flooded	94.00	62.00	74.00
Flooded	73.00	96.00	83.00
Accuracy (%)	79.38		
ROC-AUC (%)	95.36		
Fold 2			
Non-Flooded	84.00	81.00	83.00
Flooded	83.00	86.00	84.00
Accuracy (%)	83.51		
ROC-AUC (%)	90.55		
Fold 3			
Non-Flooded	97.00	65.00	78.00
Flooded	75.00	98.00	85.00
Accuracy (%)	82.29		
ROC-AUC (%)	97.65		

Table B7
Spatial cross-validation Results across All Folds for the XGBoost Model

Fold 1			
	Precision (%)	Recall (%)	f1-score (%)
Non-Flooded	62.00	55.00	58.00
Flooded	62.00	69.00	69.00
Accuracy (%)	62.32		
ROC-AUC (%)	59.60		
Fold 2			
Non-Flooded	80.00	73.00	77.00
Flooded	62.00	71.00	67.00
Accuracy (%)	72.53		
ROC-AUC (%)	79.59		
Fold 3			
Non-Flooded	91.00	61.00	73.00
Flooded	79.00	96.00	87.00
Accuracy (%)	82.31		
ROC-AUC (%)	88.93		

Table B8
Spatial cross-validation Results across All Folds for the CatBoost Model

Fold 1			
	Precision (%)	Recall (%)	f1-score (%)
Non-Flooded	59.00	73.00	65.00
Flooded	68.00	53.00	59.00
Accuracy (%)	62.32		
ROC-AUC (%)	62.29		
Fold 2			
Non-Flooded	82.00	84.00	83.00
Flooded	74.00	71.00	72.00
Accuracy (%)	79.12		
ROC-AUC (%)	86.58		
Fold 3			
Non-Flooded	72.00	75.00	73.00
Flooded	83.00	81.00	82.00
Accuracy (%)	78.46		
ROC-AUC (%)	86.62		

Table B9
Spatial cross-validation Results across All Folds for the LightGBM Model

Fold 1			
	Precision (%)	Recall (%)	f1-score (%)
Non-Flooded	56.00	70.00	62.00
Flooded	64.00	50.00	56.00
Accuracy (%)	59.42		
ROC-AUC (%)	61.11		
Fold 2			
Non-Flooded	82.00	75.00	79.00
Flooded	65.00	74.00	69.00
Accuracy (%)	74.73		
ROC-AUC (%)	80.97		
Fold 3			
Non-Flooded	97.00	61.00	75.00
Flooded	80.00	99.00	88.00
Accuracy (%)	83.85		
ROC-AUC (%)	87.54		

Table B10
Spatial cross-validation Results across All Folds for the Random Forest Model

Fold 1			
	Precision (%)	Recall (%)	f1-score (%)
Non-Flooded	51.00	73.00	60.00
Flooded	59.00	36.00	45.00
Accuracy (%)	53.62		
ROC-AUC (%)	59.76		
Fold 2			
Non-Flooded	84.00	73.00	78.00
Flooded	64.00	77.00	70.00
Accuracy (%)	74.73		
ROC-AUC (%)	82.65		
Fold 3			
Non-Flooded	93.00	73.00	81.00
Flooded	84.00	96.00	90.00
Accuracy (%)	86.92		
ROC-AUC (%)	91.02		

Table B11
Aggregated Confusion Matrices for All Models (Stratified Cross-Validation)

	XGBoost			CatBoost		
	Predicted			Predicted		
		Non-Flooded	Flooded		Non-Flooded	Flooded
Actual	Non-Flooded	99	41	Non-Flooded	113	27
	Flooded	15	135	Flooded	16	134
	LightGBM			**Random Forest**		
	Predicted			Predicted		
		Non-Flooded	Flooded		Non-Flooded	Flooded
	Non-Flooded	111	29	Non-Flooded	97	43
	Flooded	12	138	Flooded	10	140

Table B12
Aggregated Confusion Matrices for All Models (Spatial Cross-Validation)

	XGBoost			CatBoost		
	Predicted			Predicted		
		Non-Flooded	Flooded		Non-Flooded	Flooded

(*continued on next page*)

Table B12 (*continued*)

	XGBoost			CatBoost		
	Predicted			Predicted		
		Non-Flooded	Flooded		Non-Flooded	Flooded
Actual	Non-Flooded	90	50	Non-Flooded	109	31
	Flooded	24	126	Flooded	42	108
	LightGBM			**Random Forest**		
	Predicted			Predicted		
		Non-Flooded	Flooded		Non-Flooded	Flooded
	Non-Flooded	96	44	Non-Flooded	102	38
	Flooded	28	122	Flooded	34	116

Appendix C. Random Index (RI) Values, Consistency Index (CI) and Consistency Ratio (CR) Formulas, and Pairwise Comparison Matrices of Experts Used in the Study (*Related to Section 2.5.2 Expert Pairwise Comparisons)*

Table C1
AHP Random Index (RI) Values for Matrices of Various Sizes.

n	3	4	5	6	7	8	9	10
RI	0.58	0.9	1.12	1.24	1.32	1.41	1.45	1.49

Equation (C.1). Equations for Consistency Index (CI) and Consistency Ratio (CR) in AHP

$$CI = \frac{\lambda\, max - n}{n - 1} \tag{1}$$

Where:

- λ_{max}: Represents the maximum eigenvalue of the comparison matrix.
- n: Represents the number of criteria.

$$CR = \frac{CI}{RI} \tag{2}$$

Table C2
Pairwise Comparison Matrix Using Saaty Scale for Expert 1 and Derived Criteria Weights

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C1	1	2	4	3	0.33	3	2	**0.21**	**2**
C2	0.50	1	3	2	0.50	2	1	**0.14**	**3**
C3	0.25	0.33	1	0.50	0.2	0.50	0.33	**0.05**	**7**
C4	0.33	0.5	2	1	0.25	1	0.50	**0.07**	**6**
C5	3	2	5	4	1	4	2	**0.32**	**1**
C6	0.33	0.50	2	1	0.25	1	0.50	**0.07**	**5**
C7	0.50	1	3	2	0.50	2	1	**0.14**	**4**
CI	0.02								
RI	1.32								
CR	**0.02**								

In the previous table and all subsequent tables, the criteria symbols are as follows: **C1:** Slope, **C2:** Elevation, **C3:** Plan Curvature, **C4:** Depth of Depressions, **C5:** Distance to the Drainage Network, **C6:** Land use/land cover, **C7:** Precipitation.

Table C3
Pairwise Comparison Matrix Using Saaty Scale for Expert 2 and Derived Criteria Weights

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C1	1	5	3	4	5	3	2	**0.33**	**1**
C2	0.20	1	0.33	0.50	1	0.33	0.25	**0.05**	**6**
C3	0.33	3	1	2	3	1	0.50	**0.13**	**4**
C4	0.25	2	0.50	1	2	0.50	0.33	**0.08**	**5**

(*continued on next page*)

Table C3 (*continued*)

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C5	0.20	1	0.33	0.50	1	0.33	0.25	**0.05**	**7**
C6	0.33	3	1	2	3	1	0.50	**0.13**	**3**
C7	0.50	4	2	3	4	2	1	**0.22**	**2**
CI	0.02								
RI	1.32								
CR	**0.01**								

Table C4
Pairwise Comparison Matrix Using Saaty Scale for Expert 3 and Derived Criteria Weights

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C1	1	3	2	3	0.50	2	2	**0.20**	**2**
C2	0.33	1	0.50	1	0.25	0.50	0.50	**0.06**	**7**
C3	0.50	2	1	2	0.33	1	1	**0.12**	**3**
C4	0.33	1	0.50	1	0.25	0.50	0.50	**0.06**	**6**
C5	2	4	3	4	1	3	3	**0.32**	**1**
C6	0.50	2	1	2	0.33	1	1	**0.12**	**4**
C7	0.50	2	1	2	0.33	1	1	**0.12**	**5**
CI	0.007								
RI	1.32								
CR	**0.005**								

Table C5
Pairwise Comparison Matrix Using Saaty Scale for Expert 4 and Derived Criteria Weights

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C1	1	1	0.33	0.33	0.50	0.33	2	**0.08**	**5**
C2	1	1	0.33	0.33	0.50	0.33	2	**0.08**	**6**
C3	3	3	1	1	2	1	4	**0.22**	**2**
C4	3	3	1	1	2	1	4	**0.22**	**3**
C5	2	2	0.50	0.50	1	0.50	3	**0.13**	**4**
C6	3	3	1	1	2	1	4	**0.22**	**1**
C7	0.50	0.50	0.25	0.25	0.33	0.25	1	**0.05**	**7**
CI	0.007								
RI	1.32								
CR	**0.005**								

Table C6
Pairwise Comparison Matrix Using Saaty Scale for Expert 5 and Derived Criteria Weights

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C1	1	1	2	2	0.50	1	2	**0.16**	**3**
C2	1	1	2	2	0.50	1	2	**0.16**	**4**
C3	0.50	0.50	1	1	0.33	0.50	1	**0.08**	**5**
C4	0.50	0.5	1	1	0.33	0.50	1	**0.08**	**6**
C5	2	2	3	3	1	2	3	**0.28**	**1**
C6	1	1	2	2	0.50	1	2	**0.16**	**2**
C7	0.50	0.50	1	1	0.33	0.50	1	**0.08**	**7**
CI	0.002								
RI	1.32								
CR	**0.002**								

Table C7
Pairwise Comparison Matrix Using Saaty Scale for Expert 6 and Derived Criteria Weights

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C1	1	1	1	1	0.33	0.50	1	**0.10**	**3**
C2	1	1	1	1	0.33	0.50	1	**0.10**	**4**
C3	1	1	1	1	0.33	0.50	1	**0.10**	**5**
C4	1	1	1	1	0.33	0.50	1	**0.10**	**6**
C5	3	3	3	3	1	2	3	**0.31**	**1**
C6	2	2	2	2	0.50	1	2	**0.19**	**2**

(*continued on next page*)

Table C7 (*continued*)

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C7	1	1	1	1	0.33	0.50	1	**0.10**	**7**
CI	0.001								
RI	1.32								
CR	**0.001**								

Table C8
Pairwise Comparison Matrix Using Saaty Scale for Expert 7 and Derived Criteria Weights

Criteria	C1	C2	C3	C4	C5	C6	C7	Weight	Rank
C1	1	0.50	3	3	0.33	2	3	**0.15**	**3**
C2	2	1	4	4	0.50	3	4	**0.23**	**2**
C3	0.33	0.25	1	1	0.20	0.33	1	**0.05**	**5**
C4	0.33	0.25	1	1	0.2	0.33	1	**0.05**	**6**
C5	3	2	5	5	1	4	5	**0.34**	**1**
C6	0.50	0.33	3	3	0.25	1	3	**0.12**	**4**
C7	0.33	0.25	1	1	0.20	0.33	1	**0.05**	**7**
CI	0.03								
RI	1.32								
CR	**0.02**								

Appendix D. Locations of Historically Flood-Prone and Low-Risk Areas, Individual Model Results, and Integrated Scenarios *(Related to* Section 2.*8 Assessment Scenario)*

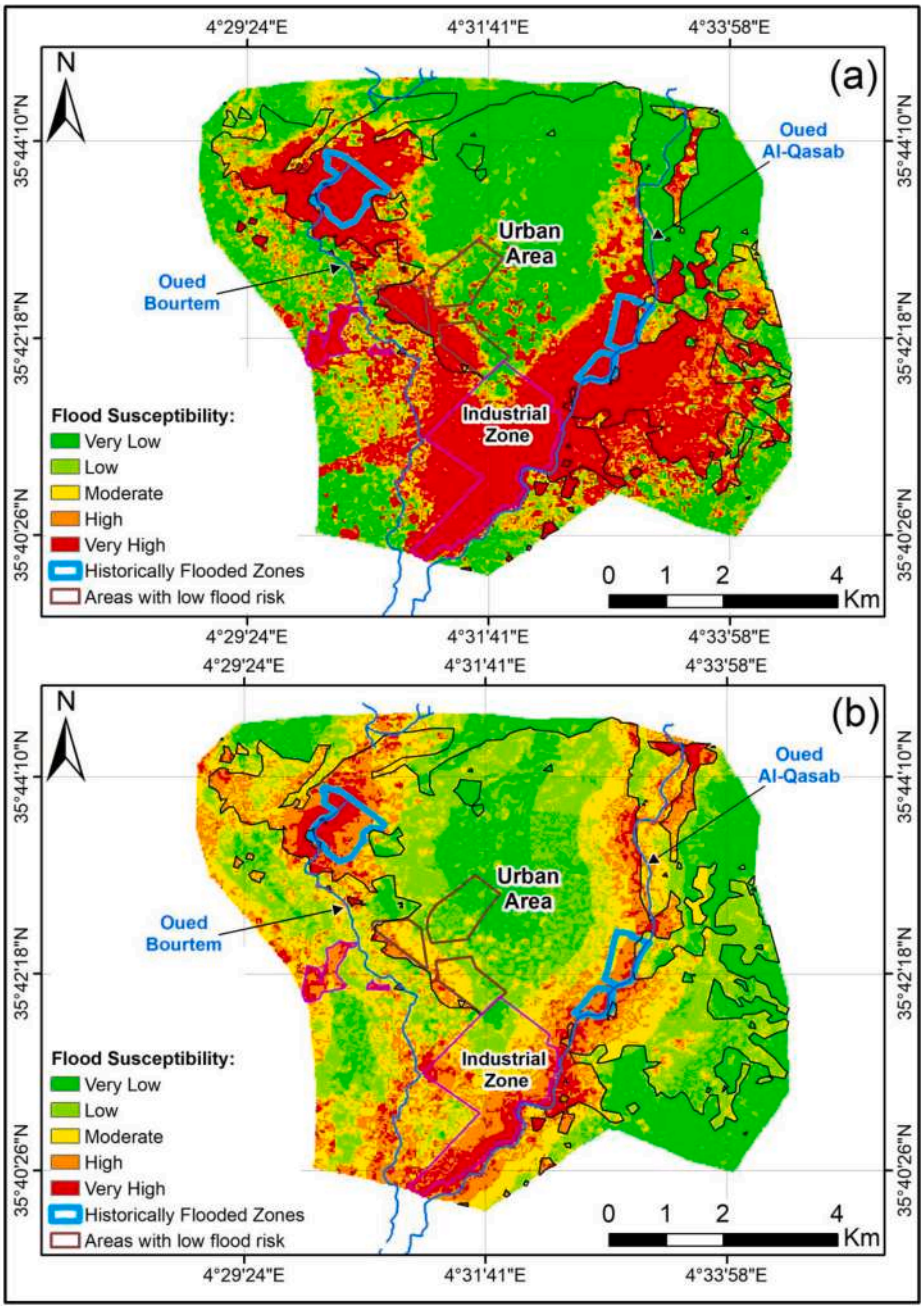


Fig. D1. Locations of Historically Flood-Prone and Low-Risk Areas with Individual Model Results: (a) CatBoost Model Output, (b) Expert-Based AHP Map.

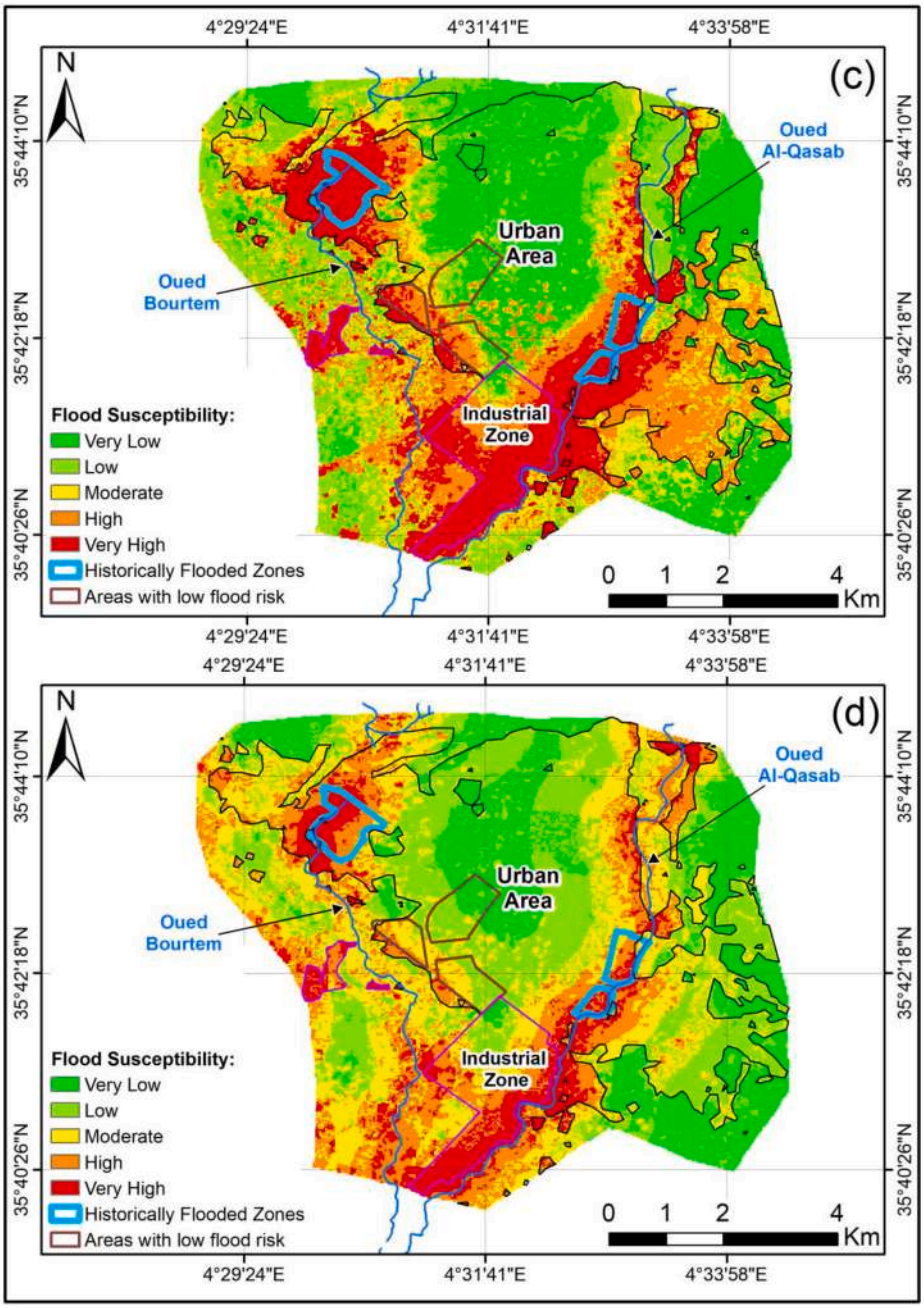


Fig. D2. Integrated Flood Susceptibility Maps Using Different Strategies: (c) Map-Level Integration (CatBoost + AHP), (d) Weight-Level Integration (AHP + SHAP).

Table D1
Flood Event Frequency and Associated Human and Material Losses in the Selected Validation Areas

Event Date	Neighborhood	Human Losses	Material Losses
10 May 1982	EL ARGOUB EL KOUCH	2 fatalities, ~30 injured	6 houses collapsed
20 October 1982	EL ARGOUB EL KOUCH	51 injured	9 houses collapsed
29 October 1982	EL ARGOUB EL KOUCH	40 injured	7 houses collapsed
23 September 1994	EL ARGOUB EL KOUCH	1 fatality, 810 homeless	256 houses collapsed
29 September 2001	EL ARGOUB EL KOUCH	Few injuries	Partial and temporary road disruption

(*continued on next page*)

Table D1 (*continued*)

Event Date	Neighborhood	Human Losses	Material Losses
23 September 2007	EL ARGOUB EL KOUCH MOUILEHA	10 fatalities, 60 homeless	200 billion centimes (estimated damages)
09 October 2013	EL ARGOUB EL KOUCH MOUILEHA	1 fatality, a few injuries	2 houses collapsed, Partial and temporary road disruption
22 January 2017	EL ARGOUB EL KOUCH The neighborhoods located in the expansion zone (POS 05)	None reported	Partial collapse of some houses, Temporary road disruption; fallen trees and electricity poles
6 June 2018	EL ARGOUB EL KOUCH	None reported	Water leakage in a mosque, Partial and temporary road disruption
03 May 2021	The neighborhoods located in the expansion zone (POS 05) EL ARGOUB EL KOUCH	A few injuries	Leakage in houses and public institutions; partial collapse of houses, Temporary road disruption; vehicles swept away by floods

Appendix E. Detailed Class Area Statistics and Percentage Change Analysis under Weight Variations *(3.5 Sensitivity Analysis Outcomes and Scenario Comparison)*

Table E1
Computed Class Areas under Weight Variations (km^2)

Scenario I: Map-Level Integration of CatBoost and AHP Results					
Class	Reference 0.6/0.4	0.4/0.6	0.5/0.5	0.7/0.3	0.3/0.7
Very Low	13.11	15.88	14.79	13.17	18.24
Low	12.79	10.35	11.67	13.34	8.43
Moderate	8.83	6.59	7.66	11.01	5.79
High	11.07	8.91	10.48	10.83	7.32
Very High	10.53	14.60	11.72	7.98	16.55
Scenario II: Weight-Level Integration of SHAP and Expert Judgments within an AHP					
Class	Reference 0.6/0.4	0.4/0.6	0.5/0.5	0.7/0.3	0.3/0.7
Very Low	10.09	10.36	10.15	9.48	10.63
Low	15.69	16.44	15.38	15.35	16.07
Moderate	14.46	13.89	13.93	14.87	13.99
High	10.78	10.25	11.23	11.14	10.27
Very High	5.35	5.43	5.68	5.55	5.41

Table E2
Relative Area Change (%) under Weight Variations (Computed Using Equation (4), Section 2.7)

Scenario I: Map-Level Integration of CatBoost and AHP Results				
Class	**0.4/0.6**	**0.5/0.5**	**0.7/0.3**	**0.3/0.7**
Very Low	21.16	12.79	0.44	39.12
Low	−19.06	−8.70	4.36	−34.03
Moderate	−25.35	−13.25	24.65	−34.43
High	−19.57	−5.32	−2.21	−33.92
Very High	38.66	11.35	−24.20	57.19
Scenario II: Weight-Level Integration of SHAP and Expert Judgments within an AHP				
Class	0.4/0.6	0.5/0.5	0.7/0.3	0.3/0.7
Very Low	2.66	0.60	−6.29	5.36
Low	4.81	−1.99	−2.14	2.42
Moderate	−3.93	−3.62	2.91	−3.23
High	−4.97	4.18	3.28	−4.76
Very High	1.51	6.04	3.67	1.12

Data availability

Data will be made available on request.

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